

Automated Goal-Based Adaptivity for Time-Dependent Partial Differential Equations



Yue Wu

Pembroke College

University of Oxford

A thesis submitted for the degree of

M.Sc. in Mathematical Modelling and Scientific Computing

Trinity Term 2026

Acknowledgements

It doesn't really matter in which order the acknowledgements and abstract come. In some ways it's nice to put the acknowledgements first then everything that follows is about the project you have done.

You should generally thank your supervisor(s) in your acknowledgements — not doing so makes quite a statement. If anyone else has helped with the project then thank them too.

If you have received funding from anywhere, it is standard to thank your funding body.

It's then up to you who else to thank. You can be brief and stop there, or you can thank family, friends, the college cat ...

Abstract

Goal-oriented adaptive finite element methods refine a discretisation to reduce the error in a prescribed quantity of interest. For time-dependent partial differential equations, their automation requires a global dual-weighted residual (DWR) estimator to be localised in both space and time. This dissertation develops an automated localisation method that operates directly on the weak variational formulation. It extends hierarchical bubble-cone recovery to discontinuous-Galerkin time discretisations and separates the residual into cell-volume, spatial-facet, temporal-interface and, when necessary, mixed space-time ridge contributions. The resulting local indicators drive a complete solve-estimate-mark-refine cycle with slabwise spatial refinement and temporal bisection.

The framework is implemented in **Firedrake** and tested on heat, advection-diffusion, Benjamin-Bona-Mahony, and parabolic p -Laplace problems. The adapted meshes agree with the expected propagation of goal sensitivity and concentrate resolution where it most affects the selected output. These experiments provide numerical evidence that the entity-wise residual recovery developed in this dissertation offers an effective route to automated goal-based space-time adaptivity.

Contents

1	Introduction	1
1.1	Motivation and Background	1
1.2	The Localisation Problem in Space–Time	2
1.3	Objectives and Main Contributions	4
1.4	Code and Software	4
1.5	Structure of the Dissertation	4
2	Mathematical and Computational Framework	6
2.1	Adaptive Finite Element Methods	6
2.2	The Dual-Weighted Residual (DWR) Method	7
2.2.1	Quantities of Interest and Adjoint Problem	7
2.2.2	An Error Identity	7
2.2.3	Computable Estimators and Effectivity	9
2.3	Error Localisation Strategies	10
2.3.1	Stationary Bubble and Cone Projection	11
2.3.2	Space–Time Extension of the Residual Recovery	12
2.3.3	Localisation of the Nonlinear Adjoint-Residual Term	16
2.3.4	Localisation Effectivity	17
2.4	Marking and Refinement	17
2.5	Automated Adaptive Algorithm	18
2.6	Code Implementation	19
2.6.1	Primal, Goal, and Adjoint Solves	19
2.6.2	Residual Localisation	20
2.6.3	Transfer on Nonmatching Slab Meshes	20
3	Experiments	21
3.1	Two-dimensional Rotating-Hill Heat Equation	21
3.1.1	Manufactured Problem and Quantity of Interest	21

3.1.2	Two Refinement Indicators	22
3.1.3	Results and Interpretation	24
3.2	One-dimensional Advection-Diffusion Equation	26
3.2.1	Moving-pulse Problem	27
3.2.2	Quantity of Interest and Adjoint Sensitivity	27
3.2.3	One-dimensional Residual Localisation	29
3.2.4	Results and Interpretation	29
3.3	Benjamin–Bona–Mahony (BBM) Equation	31
3.3.1	Primal Problem and Solitary Wave	31
3.3.2	Quantity of Interest	32
3.3.3	Localisation: Mixed Space–Time Ridge	33
3.3.4	Results and Interpretation	34
3.4	Parabolic p -Laplace Problem on an L-shaped Domain	36
3.4.1	p -Laplace Problem and Goal Functional	36
3.5	Incompressible Navier–Stokes Equations	39
4	Conclusions and Future Work	42
4.1	Summary	42
4.2	Future Work	42
4.3	Conclusion	42
4.4	Page Numbering	42
	References	43

List of Figures

2.1	Bubble and cone test functions on a single triangular element K . . .	12
2.2	Space-time cell $Q_{K,n} = K \times I_n$	14
3.1	Numerical rotating-hill solution at the four physical snapshot times $t = 0.25, 0.50, 0.75, 1.00$	25
3.2	Slabwise spatial meshes from strong-residual-bound marking at $t = 0.25, 0.50, 0.75, 1.00$ at the 8th iteration.	25
3.3	Slabwise spatial meshes from the bubble run at the physical snapshot times $t = 0.25, 0.50, 0.75, 1.00$ at the 8th iteration.	26
3.4	Exact solution of the 1D advection–diffusion problem. The dashed line indicates the nominal pulse trajectory $x = x_0 + bt$	28
3.5	Evolution of the space-time mesh for the Gaussian detector centred at $x_d = 0.8$	30
3.6	Space-time cells marked during the first adaptive iteration for Gaussian terminal sensors centred at three different locations.	30
3.7	Final adapted space-time meshes for Gaussian terminal sensors centred at $x_d = 0.7, 0.8$, and 0.9	31
3.8	Evolution of the independently refined slabwise space-time mesh. . .	35
3.9	Spatial meshes at time $t = 0, 0.5, 0.75, 1$ for goal J_1 at iteration 7. . .	37
3.10	J_1 error against total primal space-time degrees of freedom for adaptive iterations 0–7 and the uniform sequence 0–3.	38
3.11	Spatial meshes for goal J_2 at iteration 7. (a) shows the initial mesh, (b)–(e) show the spatial meshes carried by the adaptive time slabs whose right endpoints are $t = 0.5, 0.625, 0.75$, and 1 , respectively. . .	40
3.12	Stacked representation of the slabwise spatial meshes at adaptive iterate 7, in exterior view and a view through the re-entrant corner, respectively. The red prism represents the space-time observation region $Q_I = \Omega_I \times [0.5, 0.75]$	41

List of Tables

3.1	Selected adaptive iterations for the strong-residual bound and bubble localisation. Here $N_{\text{st}} = \sum_{n=1}^{N_t} \dim(\text{CG}(1; \mathcal{T}_n))$, $N_{\text{cell}} = \sum_{n=1}^{N_t} \#\mathcal{T}_n$. . .	24
3.2	Mixed-ridge ablation performed on identical discrete primal and adjoint solutions.	34
3.3	Selected iterations of the periodic BBM adaptive calculation.	35
3.4	Representative fitted- J_1 results.	38

Chapter 1

Introduction

1.1 Motivation and Background

In many partial differential equation (PDE) problems, the numerical solution is not required with uniform accuracy throughout the computational domain. What matters instead is the accurate evaluation of a specific *quantity of interest* (QoI), a functional

$$J : U \longrightarrow \mathbb{R} \tag{1.1}$$

that encodes why the simulation is being run in the first place, where U is the state space. If $u \in U$ is the exact solution and u_h is a finite element approximation, the error relevant to the stated purpose of the computation is

$$e_J := J(u) - J(u_h). \tag{1.2}$$

For an evolution problem, J may describe a terminal observation, an accumulated response over time, or a combination of the two, and it may be linear or nonlinear in the state.

Conventional adaptive finite element methods (AFEM) commonly seek to control a norm of the state error $u - u_h$ by a SOLVE–ESTIMATE–MARK–REFINE cycle [3, 1, 15, 21]. This is appropriate when the accuracy of the state throughout the computational domain is itself the objective. However, when the aim is to compute only a selected output, the relevant question is not where the numerical error is largest, but where it has the largest effect on that output. Refinement based only on a global state norm may therefore spend computational effort in regions that have little influence on J . Goal-oriented adaptivity instead seeks to identify and reduce the discretisation errors that matter for the chosen functional.

The dual-weighted residual (DWR) method provides a systematic framework for goal-oriented error estimation [5, 6, 4]. It combines the residual of the discrete primal solution with a solution of an adjoint problem. The residual measures the defect of the numerical state in the variational equation, while the adjoint measures the sensitivity of the QoI to that defect. Because the adjoint is determined by J , changing the functional while keeping the primal PDE fixed can change the dual weight, the local refinement indicators, and ultimately the adapted mesh.

Exact DWR identities contain errors in the primal and adjoint solutions that are not available computationally. In practice, these terms are approximated using enriched finite element spaces. The detailed identities, computable approximations, and the assumptions under which they estimate the true goal error are presented in Chapter 2.

1.2 The Localisation Problem in Space–Time

A global DWR estimator indicates whether the selected output is sufficiently accurate, but it does not by itself determine where the discretisation should be refined. To drive adaptation, it must be decomposed into contributions associated with spatial cells and time intervals.

Classical elementwise integration by parts produces strong cell residuals, spatial flux jumps, boundary terms, and, for discontinuous time discretisations, temporal-jump terms. These contributions are interpretable for a fixed equation, but their derivation depends on the differential operator, boundary conditions, and discretisation. Repeating that derivation for every new variational problem limits automation.

Variational filtering and partition-of-unity localisation avoid an explicit strong-form derivation by acting directly on the weak residual [8, 24]. Space–time partition-of-unity DWR methods have been developed for several classes of nonstationary problems [28, 12, 26]. Their local contributions are naturally associated with basis functions or patches. Beyond these generic constructions, localisation can also exploit structure specific to a problem class or discretisation. Examples include gradient-averaged and nodal decompositions for convection–diffusion transport [19], equilibrated flux and potential reconstructions for elliptic and parabolic approximations [22, 10], space–time error splitting for dynamically changing meshes [27], and coarse-neighbourhood indicators for high-contrast multiscale models [9].

The localisation developed in this dissertation is motivated by the stationary bubble–cone construction of Rognes and Logg [25]. That construction recovers poly-

nomial residual densities by evaluating a weak residual on support-selective bubble and cone test functions. Here the same principle is extended to space–time slabs. Tensor-product spatial and temporal test functions are used in a hierarchical sequence to recover volume, spatial-facet, temporal-face, and, when nonzero, mixed-ridge contributions. At each stage, contributions already exposed on higher-dimensional entities are subtracted before the next residual density is recovered. The resulting entity contributions are paired with a computable DWR weight to obtain signed cell–slab indicators.

Time dependence introduces more than an additional coordinate. In this dissertation, time is divided into slabs and discretised by discontinuous Galerkin methods. The weak residual therefore contains both slab-interior actions and defects at temporal interfaces. Moreover, slab-local spatial refinement can give consecutive finite element spaces $V_{n-1} \neq V_n$. The outgoing primal state must then be transferred forward between meshes, and adjoint sensitivity must be propagated backwards using the adjoint of the same discrete transfer. The transferred interface defect is also part of the temporal residual and must be treated consistently by the localisation.

The geometric support of an interface residual also depends on the time-mass pairing of the evolution equation. For the standard spatial L^2 pairing, the discontinuous-Galerkin defect acts on a temporal face. If the pairing contains spatial derivatives, integration over that face can additionally reveal a contribution on the intersection of a spatial facet and a temporal interface. This lower-dimensional entity is referred to here as a mixed space–time ridge. It is not required for every problem, but it cannot in general be represented by merely increasing the polynomial degree of a volume recovery.

The stationary recovery has an established exact or projective interpretation under polynomial-representation and stability assumptions. The space–time extension proposed here is a computational construction. It is therefore assessed numerically in two distinct ways. The *effectivity index* compares the global estimator with a reference goal error, whereas the *localisation gap* measures whether the recovered local contributions reproduce the independently assembled global DWR estimator. These tests separate the quality of the enriched primal–adjoint approximation from the algebraic completeness of the localisation.

1.3 Objectives and Main Contributions

The objective of this dissertation is to develop and investigate an automated, goal-oriented, slabwise space–time adaptive finite element workflow whose local indicators are constructed from the weak variational problem rather than from a problem-specific strong-residual derivation.

A bubble–cone residual recovery is extended from stationary finite element problems to discontinuous-Galerkin time slabs. The construction distinguishes volume, spatial-facet, and temporal-face residual entities and combines them into signed cell–slab DWR contributions.

1.4 Code and Software

The implementation is written in Python using **Firedrake**, which represents and assembles finite element variational forms through the Unified Form Language (UFL) [18]. UFL differentiation is used to form the linearised and transposed spatial operators and the derivatives of the specified goal functional. The time discretisation follows the automated Galerkin time-stepping framework developed in and around **Irksome** [17, 2].

The development repository can be found at <https://github.com/yuewu2/mmsc-dissertation>.

1.5 Structure of the Dissertation

Chapter 2 develops the mathematical and computational framework. It first introduces AFEM and the linear and nonlinear DWR error representations, then describes enriched estimators and effectivity. It next reviews residual localisation and develops the stationary and space–time bubble–cone recovery, including temporal interfaces, the optional mixed ridge, temporal quadrature, and localisation diagnostics. The chapter concludes by incorporating these components into a complete slabwise adaptive algorithm and by specifying transfer between nonmatching spatial meshes.

Chapter 3 presents the numerical investigation. The rotating-hill heat equation compares automated bubble recovery with a manually derived strong-residual strategy. The advection–diffusion example demonstrates the dependence of the adapted space–time mesh on the chosen goal functional. The nonlinear BBM example then tests the estimator for a dispersive equation and uses a localisation gap to identify

the mixed space–time ridge. The final L-shaped parabolic p -Laplace example examines the nonlinear estimator in the presence of a re-entrant-corner singularity and a localised observation region. The final chapter summarises the findings, limitations, and directions for further work.

Chapter 2

Mathematical and Computational Framework

2.1 Adaptive Finite Element Methods

Let U and V be reflexive Banach spaces with dual spaces U^* and V^* , respectively. Let $\mathcal{A} : U \rightarrow V^*$ denote the weak residual operator associated with the problem. The continuous problem is written as

$$\mathcal{A}(u)(v) = 0 \quad \text{for all } v \in V. \quad (2.1)$$

For finite-dimensional spaces $U_h \subset U$ and $V_h \subset V$, a Galerkin approximation $u_h \in U_h$ satisfies

$$\mathcal{A}(u_h)(v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (2.2)$$

The development of adaptive finite element methods (AFEM) was inspired by the pioneering work of Babuška and Rheinboldt [3]. To make an adaptive solver practically realisable, and to study the optimality of adaptive procedures, the basic AFEM algorithm is organised as a SOLVE–ESTIMATE–MARK–REFINE loop [14]:

1. SOLVE the differential equation on the current space–time discretisation.
2. ESTIMATE global estimator for the error quantity under consideration.
3. MARK the elements by localising the error estimator.
4. REFINE the marked elements using a specified refinement strategy.

The automated goal-oriented solver considered in this dissertation follows this AFEM framework. The dual-weighted residual method, introduced in the following

section, provides the goal-oriented error estimator used in the ESTIMATE step. Section 2.3 describes how this global estimator is decomposed into local contributions, while the marking and refinement strategy is specified at the end of this chapter.

2.2 The Dual-Weighted Residual (DWR) Method

2.2.1 Quantities of Interest and Adjoint Problem

As introduced in (1.1)–(1.2), the quantity of interest may be linear or nonlinear and may contain spatial-volume, boundary, running-in-time, or terminal observations. We now make its role in the adjoint problem precise.

The DWR method requires an adjoint problem determined by the chosen functional J . With primes denoting Fréchet derivatives, the continuous adjoint is: find $z \in V$ such that

$$\mathcal{A}'(u)(w, z) = J'(u)(w) \quad \text{for all } w \in U, \quad (2.3)$$

where u solves (2.1). The corresponding discrete adjoint is: find $z_h \in V_h$ such that

$$\mathcal{A}'(u_h)(w_h, z_h) = J'(u_h)(w_h) \quad \text{for all } w_h \in U_h. \quad (2.4)$$

where u_h solves (2.2).

2.2.2 An Error Identity

Theorem 2.2.1 (Nonlinear DWR identity; [6, 14]). *Assume that $\mathcal{A} \in C^3(U, V^*)$ and $J \in C^3(U, \mathbb{R})$. For any approximation $\tilde{x} = (\tilde{u}, \tilde{z}) \in U \times V$,*

$$J(u) - J(\tilde{u}) = \frac{1}{2}\rho(\tilde{u})(z - \tilde{z}) + \frac{1}{2}\rho^*(\tilde{u}, \tilde{z})(u - \tilde{u}) - \rho(\tilde{u})(\tilde{z}) + \mathcal{R}^{(3)}, \quad (2.5)$$

where

$$\rho(\tilde{u})(\cdot) := -\mathcal{A}(\tilde{u})(\cdot), \quad (2.6)$$

$$\rho^*(\tilde{u}, \tilde{z})(\cdot) := J'(\tilde{u})(\cdot) - \mathcal{A}'(\tilde{u})(\cdot, \tilde{z}). \quad (2.7)$$

Moreover, with $e = u - \tilde{u}$ and $e^* = z - \tilde{z}$,

$$\begin{aligned} \mathcal{R}^{(3)} = \frac{1}{2} \int_0^1 & \left[J'''(\tilde{u} + se)(e, e, e) - \mathcal{A}'''(\tilde{u} + se)(e, e, e, \tilde{z} + se^*) \right. \\ & \left. - 3\mathcal{A}''(\tilde{u} + se)(e, e, e^*) \right] s(s-1) ds. \end{aligned} \quad (2.8)$$

Proof sketch. Set $X = U \times V$, $x = (u, z)$, $\tilde{x} = (\tilde{u}, \tilde{z})$, and introduce the Lagrangian

$$\mathcal{L}(\hat{u}, \hat{z}) := J(\hat{u}) - \mathcal{A}(\hat{u})(\hat{z}). \quad (2.9)$$

Along the segment $x_s = \tilde{x} + s(x - \tilde{x})$, the fundamental theorem of calculus and the trapezoidal identity give

$$\mathcal{L}(x) - \mathcal{L}(\tilde{x}) = \frac{1}{2}[\mathcal{L}'(x)(x - \tilde{x}) + \mathcal{L}'(\tilde{x})(x - \tilde{x})] + \frac{1}{2} \int_0^1 \mathcal{L}'''(x_s)(x - \tilde{x})^3 s(s-1) ds. \quad (2.10)$$

The exact primal and adjoint equations make x a stationary point of $\mathcal{L}$, so $\mathcal{L}'(x) = 0$. Expanding $\mathcal{L}'(\tilde{x})$ in its primal and dual directions produces the primal residual ρ and adjoint residual ρ^* ; substituting the definition of $\mathcal{L}$ gives the Galerkin correction in (2.5). Expanding the third derivative gives (2.8). Complete algebraic details are given in [23, 13, 14]. $\square$

The correction term $-\rho(\tilde{u})(\tilde{z})$ vanishes when $\tilde{u} = u_h$ is solved to Galerkin accuracy and $\tilde{z} \in V_h$. If $\mathcal{A}$ is affine and J is linear, then $\mathcal{R}^{(3)} = 0$, and the two half-residual terms coincide. To state the resulting linear identity explicitly, let $a : U \times V \rightarrow \mathbb{R}$ be bilinear, let $\ell \in V^*$, and define

$$a(u, v) = \ell(v) \quad \forall v \in V, \quad (2.11)$$

$$a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h, \quad (2.12)$$

$$a(w, z) = J(w) \quad \forall w \in U, \quad (2.13)$$

$$\rho(u_h)(v) := \ell(v) - a(u_h, v).$$

Proposition 2.2.2 (Exact linear goal-error identity). *Let $u \in U$ solve (2.11) and $u_h \in U_h$ solve (2.12), with dual solution z defined by (2.13). Then for every $\varphi_h \in V_h$,*

$$J(u) - J(u_h) = \rho(u_h)(z - \varphi_h). \quad (2.14)$$

Proof. Writing $e = u - u_h$, the primal and adjoint equations give $\rho(u_h)(z) = a(e, z) = J(e) = J(u) - J(u_h)$. Galerkin orthogonality gives $\rho(u_h)(\varphi_h) = 0$, and subtracting this zero term proves (2.14). $\square$

The residual measures the defect of u_h in the continuous variational problem. The factor $z - \varphi_h$ is the part of the QoI sensitivity that is not represented in the discrete test space.

2.2.3 Computable Estimators and Effectivity

Both identities above contain the unknown exact solution or adjoint. In practice these are replaced by solutions in enriched spaces, obtained, for example, by uniform h -refinement, p -enrichment, or a higher-order time approximation [6, 14].

Let $U_h \subset U_h^{(2)} \subset U$ and $V_h \subset V_h^{(2)} \subset V$. The enriched primal and adjoint problems are

$$\mathcal{A}(u_h^{(2)})(v_h^{(2)}) = 0 \quad \text{for all } v_h^{(2)} \in V_h^{(2)}, \quad (2.15)$$

$$\mathcal{A}'(u_h^{(2)})(w_h^{(2)}, z_h^{(2)}) = J'(u_h^{(2)})(w_h^{(2)}) \quad \text{for all } w_h^{(2)} \in U_h^{(2)}. \quad (2.16)$$

The precise statement needed here is the enriched-space identity corresponding to Theorem 2 of [14].

Theorem 2.2.3 (Enriched-space error representation). *Suppose that restrictions $\mathcal{A}^{(2)}$ and J_h are three-times continuously differentiable on the enriched spaces and agree there with $\mathcal{A}$, $\mathcal{A}'$, J , and J' ; explicitly, for $v, w \in U_h^{(2)}$ and $\phi \in V_h^{(2)}$,*

$$\mathcal{A}(v)(\phi) = \mathcal{A}^{(2)}(v)(\phi), \quad \mathcal{A}'(v)(w, \phi) = (\mathcal{A}^{(2)})'(v)(w, \phi), \quad (2.17)$$

$$J(v) = J_h(v), \quad J'(v)(w) = J'_h(v)(w). \quad (2.18)$$

If $u_h^{(2)}$ and $z_h^{(2)}$ solve (2.15)–(2.16), then for every fixed $(\tilde{u}, \tilde{z}) \in U_h^{(2)} \times V_h^{(2)}$,

$$J(u) - J(\tilde{u}) = J(u) - J(u_h^{(2)}) + \eta^{(2)}, \quad (2.19)$$

$$\eta^{(2)} := \frac{1}{2}\rho(\tilde{u})(z_h^{(2)} - \tilde{z}) + \frac{1}{2}\rho^*(\tilde{u}, \tilde{z})(u_h^{(2)} - \tilde{u}) - \rho(\tilde{u})(\tilde{z}) + \mathcal{R}_h^{(3)}. \quad (2.20)$$

Here $\mathcal{R}_h^{(3)}$ is (2.8) with the exact errors replaced by enriched errors.

Proof. The agreement assumptions imply that $u_h^{(2)}$ and $z_h^{(2)}$ form a stationary primal–adjoint pair for the restricted Lagrangian on $U_h^{(2)} \times V_h^{(2)}$. Applying Theorem 2.2.1 on these finite-dimensional spaces yields $J_h(u_h^{(2)}) - J_h(\tilde{u}) = \eta^{(2)}$. Since $J_h = J$ on the enriched space, adding and subtracting $J(u_h^{(2)})$ gives (2.19). $\square$

Under the saturation assumption

$$|J(u) - J(u_h^{(2)})| \leq b_h |J(u) - J(\tilde{u})|, \quad 0 < b_h < 1, \quad (2.21)$$

the review [14] proves

$$\frac{1}{1+b_h} |\eta^{(2)}| \leq |J(u) - J(\tilde{u})| \leq \frac{1}{1-b_h} |\eta^{(2)}|. \quad (2.22)$$

The saturation assumption is problem- and goal-dependent and is not proved for the new space–time recovery proposed here; the experiments therefore report effectivities and localisation closure as numerical evidence.

Let $z_h \in V_h$ denote the numerical solution of the discrete dual problem on the current discretisation, and let $z_h^+ \in V_h^+$ be the corresponding dual solution computed in an enriched space $V_h^+ \supset V_h$. For a linear problem, a computable DWR estimator is

$$\eta = \rho(u_h)(z_h^+ - z_h). \quad (2.23)$$

For the dual weight $z - \varphi_h$, the literature [14] also distinguishes three reconstruction choices: a globally computed higher-order dual $\widehat{z}_h - I_h \widehat{z}_h$; a local patchwise higher-order reconstruction $I_h^+ z_h - z_h$; and a difference-quotient or flux-jump surrogate. The first is usually the most direct route to a global estimator but requires an enriched dual solve. The latter two can be cheaper for marking, but their accuracy depends on the reconstruction and is not interchangeable with that of an enriched global dual. In the present code, $z_h^+ - z_h$ and $z_h^+ - I_h z_h^+$ are both implemented. The experiments use the first choice. The extra low-order adjoint solve is more expensive than a purely local interpolation, but it gives a transparent comparison of two numerical adjoints(?). The interpolant choice remains available for a cheaper study.

When a sufficiently accurate reference value for $J(u)$ is available, the effectivity index is defined by

$$I_{\text{eff}} = \frac{\eta}{J(u) - J(u_h)}. \quad (2.24)$$

An effectivity index close to one indicates good agreement in the computed case [4].

2.3 Error Localisation Strategies

Section 1.2 motivated a decomposition of the global DWR estimator into computable local contributions. Elementwise integration by parts (IBP) gives strong cell residuals, flux jumps, and boundary residuals, but its derivation depends on the differential operator and boundary conditions. Partition-of-unity (PU) localisation inserts functions $\{\chi_i\}_{i=1}^M$, with $\sum_i \chi_i = 1$, into the weak residual and associates the resulting terms with basis functions or patches [24]. This is effective for stationary and space–time adaptivity [28, 12, 26], but a single patch contribution may combine several residual mechanisms.

The present construction pursues a more geometrically resolved decomposition. It probes the weak residual with polynomial functions supported on individual cells,

spatial facets, temporal faces, and, when present, mixed space–time ridges. The resulting coefficients separate the residual mechanisms according to their geometric support, while still being recovered directly from UFL evaluations of the weak residual in **Firedrake**. Thus no problem-specific strong residual or hand-derived flux-jump formula is needed. This approach is motivated by the stationary bubble–cone construction of Rognes and Logg [25], and is extended below to discontinuous Galerkin discretisations in time.

2.3.1 Stationary Bubble and Cone Projection

We first recall the stationary construction for the primal residual. The nonlinear adjoint-residual term is treated in Section 2.3.3. For each $K \in \mathcal{T}_h$, let $\mathcal{F}(K)$ denote its facets. Rognes and Logg formulate the following assumptions [25]:

(A1) Global decomposition. The global residual is a sum of cell-local functionals,

$$\rho(v) = \sum_{K \in \mathcal{T}_h} \rho_K(v). \quad (2.25)$$

(A2) Local decomposition. Each local residual has the form

$$\rho_K(v) = (R_K, v)_K + \sum_{F \in \mathcal{F}(K)} (R_{K,F}, v)_F. \quad (2.26)$$

(A3) Polynomial representation. The residual densities are piecewise polynomial. For chosen degrees p, q ,

$$R_K \in \mathbb{P}^p(K), \quad R_{K,F} \in \mathbb{P}^q(F). \quad (2.27)$$

Assumptions (A1) and (A2) identify the residual-supporting entities, while (A3) determines whether the recovery is exact or a weighted projection.

For a d -simplex K with vertices $\{v_i\}_{i=1}^{d+1}$ and associated barycentric coordinates $\{\lambda_i^K\}_{i=1}^{d+1}$. The cell bubble is defined by

$$b_K := \prod_{i=1}^{d+1} \lambda_i^K, \quad b_K|_{\partial K} = 0. \quad (2.28)$$

The bubble test eliminates all facet actions, so $R_K^B \in \mathbb{P}^p(K)$ can be recovered from

$$(R_K^B, b_K \phi)_K = \rho_K(b_K \phi) \quad \forall \phi \in \mathbb{P}^p(K). \quad (2.29)$$

For $F \in \mathcal{F}(K)$, let

$$\mathcal{I}_F := \{i : v_i \in F\}, \quad \beta_{K,F} := \prod_{i \in \mathcal{I}_F} \lambda_i^K. \quad (2.30)$$

This cone function is nonzero on the relative interior of F , vanishes on every other facet, and remains nonzero inside K . Let $E_{K,F} : \mathbb{P}^q(F) \longrightarrow \mathbb{P}^q(K)$, $(E_{K,F}\psi)|_F = \psi$, and write $\tilde{\psi} = E_{K,F}\psi$. Once the cell density R_K^B is recovered, the facet residual contribution $R_{K,F}^B \in \mathbb{P}^q(F)$ is defined by

$$(R_{K,F}^B, (\beta_{K,F}|_F)\psi)_F = \rho_K(\beta_{K,F}\tilde{\psi}) - (R_K^B, \beta_{K,F}\tilde{\psi})_K \quad \forall \psi \in \mathbb{P}^q(F). \quad (2.31)$$

The subtraction removes the cell action seen by the cone. Under (A1)–(A3), these weighted mass problems uniquely recover R_K and $R_{K,F}$ [25]. Figure 2.1 illustrates two representative bubble and cone test functions.



Figure 2.1: Bubble and cone test functions on a single triangular element K .

The stationary analysis is established in [25], while the space–time extension below is assessed numerically in Chapter 3 through localisation gaps, effectivities, and adaptive behaviour. The essential new feature is the dG action at each temporal interface.

2.3.2 Space–Time Extension of the Residual Recovery

Let V be a spatial trial and test space. We now consider an evolution problem of the form

$$m(\partial_t u(t), v) + a(u(t); v, t) = \ell(v; t) \quad \forall v \in V, \quad t \in (0, T), \quad (2.32)$$

with $u(0) = u_0$. Here m is the time-derivative pairing, a contains the remaining spatial and possibly nonlinear terms, and ℓ is the load. Let

$$I_n = (t_{n-1}, t_n], \quad k_n = t_n - t_{n-1}, \quad Q_{K,n} = K \times I_n, \quad \tau = \frac{t - t_{n-1}}{k_n}. \quad (2.33)$$

On I_n , trial and test functions are time polynomials with values in a spatial finite-element space V_n , and may be discontinuous at t_n . Following the standard space–time dG notation [4], we define

$$U^{n+} := \lim_{t \downarrow t_n} U(t), \quad U^{n-} := \lim_{t \uparrow t_n} U(t), \quad [U]^n := U^{n+} - U^{n-}. \quad (2.34)$$

For matching spatial spaces, the global dG semilinear form can be written as

$$\begin{aligned} \mathcal{A}(U)(V) &:= \sum_{n=1}^N \int_{I_n} [m(\partial_t U, V) + a(U; V, t) - \ell(V; t)] dt \\ &\quad + \sum_{n=2}^N m([U]^{n-1}, V^{(n-1)+}) + m(U^{0+} - u_0, V^{0+}). \end{aligned} \quad (2.35)$$

The discrete problem is $\mathcal{A}(U_h)(V_h) = 0$ for all discrete tests. Throughout this chapter we use the convention

$$\rho(U)(V) := -\mathcal{A}(U)(V). \quad (2.36)$$

The extension of the stationary recovery is based on the following assumptions.

(A1) Global decomposition. The global residual is the sum of local space–time functionals,

$$\rho(U_h)(V) = \sum_{n=1}^N \sum_{K \in \mathcal{T}_n} \rho_{K,n}(U_h)(V). \quad (2.37)$$

To be more specific, we can define the incoming defect by

$$\delta_{n-1}(U_h) := \begin{cases} U_h^{0+} - u_0, & n = 1, \\ [U_h]^{n-1}, & n > 1. \end{cases} \quad (2.38)$$

Thus

$$\begin{aligned} \rho_{K,n}(U_h)(V) &:= \int_{I_n} [\ell_{K,n}(V; t) - m_{n,K}(\partial_t U_h, V) - a_{K,n}(U_h; V, t)] dt \\ &\quad - m_{n,K}(\delta_{n-1}(U_h), V(t_{n-1}^+)). \end{aligned} \quad (2.39)$$

Note that if $V_{n-1} \neq V_n$, the outgoing trace must first be transferred into V_n :

$$\delta_{n-1}(U_h) = U_h^{(n-1)+} - P_{n-1} U_h^{(n-1)-}, \quad n > 1, \quad (2.40)$$

where $P_{n-1} : V_{n-1} \rightarrow V_n$. The transfer and its adjoint are discussed in more details in Section 2.6.3.

(A2) Local decomposition. On each space–time cell $Q_{K,n}$, the residual admits a representation by contributions supported on the cell interior, spatial facets, temporal facets, and, when required by the chosen pairing, their intersections, as shown in Figure 2.2.

$$\begin{aligned} \rho_{K,n}(U_h)(V) = & (R_{K,n}^{B,\text{vol}}, V)_{Q_{K,n}} + \sum_{F \subset \partial K} (R_{K,F,n}^{B,\text{sf}}, V)_{F \times I_n} \\ & + (R_{K,n-1}^{B,\text{tf}}, V(t_{n-1}^+))_K + \sum_{F \subset \partial K} (R_{F,n-1}^{B,\text{xt}}, V(t_{n-1}^+))_F. \end{aligned} \quad (2.41)$$

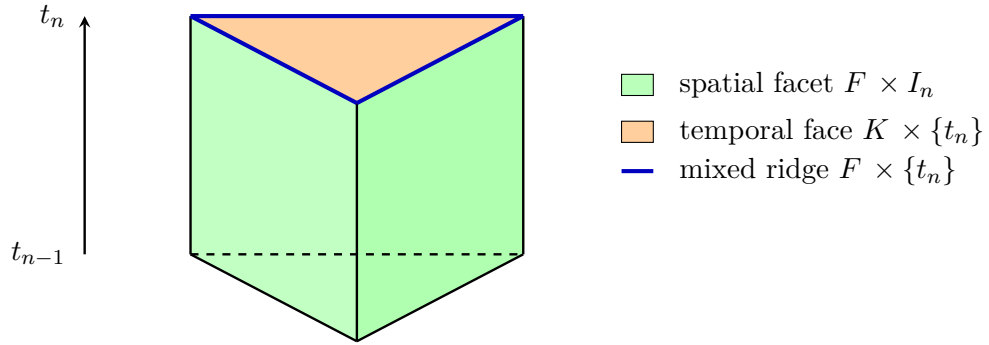


Figure 2.2: Space–time cell $Q_{K,n} = K \times I_n$.

(A3) Polynomial representation. The residual densities belong to the selected recovery spaces,

$$\begin{aligned} R_{K,n}^{\text{vol}} & \in Q_{K,n}^{p_x, p_t}, & R_{K,F,n}^{\text{sf}} & \in Q_{F,n}^{q_x, q_t}, \\ R_{K,n-1}^{\text{tf}} & \in \mathbb{P}^{r_x}(K), & R_{K,F,n-1}^{\text{xt}} & \in \mathbb{P}^{s_x}(F), \end{aligned} \quad (2.42)$$

whenever the corresponding densities are present.

Assumption (A1) follows naturally when the weak form is assembled slab by slab from cell and facet integrals. (A2) records the geometric supports. Under (A3), problems recover a density exactly when it belongs to the selected polynomial space.

Volume recovery. We first define the temporal interior bubble and its tensor product with the spatial cell bubble from eq. (2.28) by

$$b_n^t(\tau) := 4\tau(1 - \tau), \quad B_{K,n} := b_K b_n^t. \quad (2.43)$$

The function $B_{K,n}$ vanishes on every codimension-one face of $Q_{K,n}$. In particular,

$$B_{K,n}(t_{n-1}^+) = B_{K,n}(t_n^-) = 0,$$

so the temporal-interface term in eq. (2.39) vanishes under this test.

The recovered volume density $R_{K,n}^{B,\text{vol}} \in Q_{K,n}^{p_x, p_t}$ is defined by

$$(R_{K,n}^{B,\text{vol}}, B_{K,n} \Phi)_{Q_{K,n}} = \rho_{K,n}(U_h)(B_{K,n} \Phi) \quad \forall \Phi \in Q_{K,n}^{p_x, p_t}. \quad (2.44)$$

All spatial-facet and temporal-endpoint contributions vanish under this test.

Spatial-facet recovery. For $F \subset \partial K$, define

$$C_{K,F,n}^{\text{sf}} := \beta_{K,F} b_n^t. \quad (2.45)$$

Let $E_{K,F,n} : Q_{F,n}^{q_x, q_t} \rightarrow Q_{K,n}^{q_x, q_t}$ be a fixed polynomial extension and write $\tilde{\Psi} := E_{K,F,n} \Psi$. After the volume density has been recovered, we can define $R_{K,F,n}^{B,\text{sf}} \in Q_{F,n}^{q_x, q_t}$ by

$$\begin{aligned} & \left(R_{K,F,n}^{B,\text{sf}}, C_{K,F,n}^{\text{sf}} \Big|_{F \times I_n} \Psi \right)_{F \times I_n} \\ &= \rho_{K,n}(U_h)(C_{K,F,n}^{\text{sf}} \tilde{\Psi}) - \left(R_{K,n}^{B,\text{vol}}, C_{K,F,n}^{\text{sf}} \tilde{\Psi} \right)_{Q_{K,n}} \quad \forall \Psi \in Q_{F,n}^{q_x, q_t}. \end{aligned}$$

Temporal-face recovery. To isolate the incoming dG interface contribution, we define

$$C_{K,n}^{\text{tf},-} := b_K(1 - \tau), \quad (2.46)$$

which satisfies

$$C_{K,n}^{\text{tf},-}(t_{n-1}^+) = b_K, \quad C_{K,n}^{\text{tf},-}(t_n^-) = 0, \quad C_{K,n}^{\text{tf},-} \Big|_{\partial K \times I_n} = 0.$$

For $\chi \in \mathbb{P}^{r_x}(K)$, define the temporal-face test function

$$\mathcal{E}_{K,n}^{\text{tf}}(\chi) := b_K(1 - \tau)\chi. \quad (2.47)$$

which vanishes on every spatial facet and the outgoing temporal face $K \times \{t_n\}$ of $Q_{K,n}$. Hence, the local residual decomposition reduces to

$$\rho_{K,n}(U_h)(\mathcal{E}_{K,n}^{\text{tf}}(\chi)) = \left(R_{K,n}^{\text{vol}}, \mathcal{E}_{K,n}^{\text{tf}}(\chi) \right)_{Q_{K,n}} + \left(R_{K,n-1}^{\text{tf}}, b_K \chi \right)_K. \quad (2.48)$$

Once the recovered volume density $R_{K,n}^{B,\text{vol}}$ is available, define the recovered incoming temporal-face density $R_{K,n-1}^{B,\text{tf}} \in \mathbb{P}^{r_x}(K)$ by

$$\begin{aligned} (R_{K,n-1}^{B,\text{tf}}, b_K \chi)_K &= \rho_{K,n}(U_h)(\mathcal{E}_{K,n}^{\text{tf}}(\chi)) \\ &\quad - \left(R_{K,n}^{B,\text{vol}}, \mathcal{E}_{K,n}^{\text{tf}}(\chi) \right)_{Q_{K,n}}, \quad \forall \chi \in \mathbb{P}^{r_x}(K). \end{aligned} \quad (2.49)$$

Possible mixed spatial–temporal ridge. A separate mixed recovery is required only if the complete residual contains a nonzero functional supported on $F \times \{t_{n-1}\}$. Such a contribution does not normally arise from an L^2 time-mass pairing, but it can occur when the pairing contains spatial derivatives, as discussed for the BBM problem in Section 3.3.3.

For $\psi \in \mathbb{P}^{s_x}(F)$, let $\tilde{\psi} = E_{K,F}\psi$ be the fixed extension used in the stationary facet recovery and define

$$\begin{aligned} C_{K,F,n}^{\text{xt},-} &:= \beta_{K,F}(1 - \tau), \\ \mathcal{E}_{K,F,n}^{\text{xt}}(\psi) &:= C_{K,F,n}^{\text{xt},-} \tilde{\psi}. \end{aligned} \quad (2.50)$$

The mixed density $R_{K,F,n-1}^{B,\text{xt}} \in \mathbb{P}^{s_x}(F)$ is consequently defined by

$$\begin{aligned} (R_{K,F,n-1}^{B,\text{xt}}, (\beta_{K,F}|_F)\psi)_F &= \rho_{K,n}(U_h)(\mathcal{E}_{K,F,n}^{\text{xt}}(\psi)) - \left(R_{K,n}^{B,\text{vol}}, \mathcal{E}_{K,F,n}^{\text{xt}}(\psi)\right)_{Q_{K,n}} \\ &\quad - \left(R_{K,F,n}^{B,\text{sf}}, \mathcal{E}_{K,F,n}^{\text{xt}}(\psi)|_{F \times I_n}\right)_{F \times I_n} \\ &\quad - \left(R_{K,n-1}^{B,\text{tf}}, \mathcal{E}_{K,F,n}^{\text{xt}}(\psi)(t_{n-1}^+)\right)_K, \quad \forall \psi \in \mathbb{P}^{s_x}(F). \end{aligned} \quad (2.51)$$

This subtraction leaves only the contribution supported on $F \times \{t_{n-1}\}$.

2.3.3 Localisation of the Nonlinear Adjoint-Residual Term

For the nonlinear identity, the implemented estimator is

$$\eta = \frac{1}{2}\rho(u_h)(z_h^+ - z_h) + \frac{1}{2}\rho^*(u_h, z_h)(u_h^+ - u_h) - \rho(u_h)(z_h), \quad (2.52)$$

with the unknown cubic remainder omitted. The construction above recovers densities for the primal residual $\rho(u_h)$, thus it can be used directly for both primal-residual contributions and the correction term. Once u_h and z_h are fixed, the adjoint-residual term $\rho^*(u_h, z_h)(\cdot)$ is a linear functional of the primal variation. It can therefore be evaluated on the same test functions used above, but it requires a separate recovery of the linearised PDE and goal derivatives rather than reuse of the primal-residual densities.

In the present implementation, we do not perform a second bubble–cone recovery for the adjoint residual. Instead, we evaluate the adjoint-residual functional using the cellwise-constant dG0 basis functions. As these functions form a partition of unity, summing these contributions recovers the assembled adjoint-residual term. These cellwise contributions are combined with the recovered primal-residual and Galerkin-correction contributions to form the nonlinear indicator. A complete bubble–cone recovery of ρ^* remains a possible extension when a separate entity-resolved interpretation of the adjoint-residual term is needed.

2.3.4 Localisation Effectivity

Let

$$\eta_{\text{loc}} := \sum_{n,K} \eta_{K,n}, \quad (2.53)$$

To assess the quality of localisation, we define the localisation effectivity index

$$I_{\text{eff}}^{\text{loc}} := \frac{\eta_{\text{loc}}}{\eta_{\text{glob}}}, \quad (2.54)$$

whenever $\eta_{\text{glob}} \neq 0$. Exact localisation gives $I_{\text{eff}}^{\text{loc}} = 1$.

[Should I mention the scope of this method?](#)

2.4 Marking and Refinement

Let $\eta_{K,n}$ denote the signed indicator associated with the spatial cell $K \in \mathcal{T}_n$ on the time slab I_n . Each pair (K, n) is treated as one space–time marking entity. Under the default marking score, its non-negative activity is defined by

$$s_{K,n} := |\eta_{K,n}|, \quad S := \sum_{n=1}^N \sum_{K \in \mathcal{T}_n} s_{K,n}. \quad (2.55)$$

A single global Dörfler-type bulk marking operation [11] is applied to all space–time cell–slab pairs. The activities from all slabs are concatenated and sorted in non-increasing order. The marking set $\mathcal{M}_{\text{st}}$ is the smallest descending prefix whose cumulative activity reaches the prescribed fraction θ :

$$\sum_{(K,n) \in \mathcal{M}_{\text{st}}} s_{K,n} \geq \theta S. \quad (2.56)$$

The global marking set is then restricted to each time slab:

$$M_n := \{K \in \mathcal{T}_n : (K, n) \in \mathcal{M}_{\text{st}}\}. \quad (2.57)$$

Under the default independent-slab policy, only the cells in M_n are refined on I_n :

$$\mathcal{T}_n^{\text{new}} := \text{refine}(\mathcal{T}_n, M_n). \quad (2.58)$$

Temporal refinement is then controlled by a separate concentration trigger. A slab containing at least one marked cell is bisected when

$$\frac{|M_n|}{|\mathcal{T}_n|} \geq \varphi_t. \quad (2.59)$$

When I_n is bisected, both temporal children inherit $\mathcal{T}_n^{\text{new}}$.

The complete procedure therefore consists of one global space–time bulk selection, followed by slab-local spatial refinement and a separate slabwise decision on temporal bisection.

more common way in space–time adaptivity is [4]. spatial or temporal indicator: dual weight on different directions... (future work)

2.5 Automated Adaptive Algorithm

The complete solve–estimate–mark–refine cycle is summarised in Algorithm 1.

Algorithm 1 Goal-oriented slabwise space–time adaptive cycle

Require: PDE, initial slab meshes, tolerance TOL, goal functional J

- 1: **while** the stopping criterion is not satisfied **do**
 - 2: Build V_n , the enriched spaces, P_n , and P_n^* on the current slab meshes.
 - 3: Solve the primal problem forward and retain every dG slab polynomial.
 - 4: **if** (2.52) is used **then**
 - 5: Solve the enriched primal problem.
 - 6: **end if**
 - 7: Solve the low-order and enriched adjoints backward, differentiating the PDE and the actual terminal/running goal functional, and applying P_n^* at mesh changes.
 - 8: Form $z^* = z_h^+ - z_h$, or optionally $z^* = z_h^+ - I_h z_h^+$.
 - 9: Assemble the independent global DWR estimator.
 - 10: Recover volume, spatial-facet, temporal-face, and, when nonzero, mixed-ridge residuals in hierarchical order.
 - 11: Weight the recovered entities, assemble indicators $\eta_{K,n}$, and record the localisation gap (2.53).
 - 12: **if** $|\eta_{\text{glob}}| \leq \text{TOL}$ **then**
 - 13: **return** the current discretisation and estimators.
 - 14: **end if**
 - 15: Apply the marking criteria (2.56) and (2.59).
 - 16: Refine the selected slab meshes, bisect selected time intervals, and rebuild the transfer operators.
 - 17: **end while**
-

2.6 Code Implementation

2.6.1 Primal, Goal, and Adjoint Solves

The time-dependent problems are supplied as semidiscrete UFL residuals. Firedrake provides the spatial finite-element spaces and variational solvers, while Irksome’s `TimeStepper` uses these residuals to construct and solve the coupled stage systems associated with the chosen discontinuous Galerkin time discretisation [2].

Using `pyadjoint` and discard the current ufl-generated adjoint solver would be a better idea for automation. Should I rerun the experiments and so only `pyadjoint` would be included here? or maybe I’ll list this as a future work that will be carried out soon

Most of the experiments use an explicit UFL-generated adjoint implementation. In compact form, the discrete adjoint satisfies

$$\mathcal{F}'_h(U_h)^* Z_h = J'_h(U_h).$$

`UFL derivative` constructs the linearisation of the primal residual, `adjoint` transposes the resulting bilinear form, and `action` inserts the current adjoint coefficient. For nonlinear problems, the resulting form is evaluated at the saved primal polynomial at the corresponding physical time. The common solver supplies the reverse-time mass term and solves the resulting linear problem slab by slab using the same Irksome dG machinery.

Terminal and running contributions to the goal functional are handled separately because they enter the reverse-time problem differently. The derivative of a terminal goal determines the final adjoint state, whose sensitivity is subsequently propagated through all preceding slabs. The derivative of a running goal instead acts as a source throughout the backward solve. Both derivatives are generated symbolically from the user-supplied UFL goal forms.

A fully automated alternative is provided by `firedrake-adjoint`, following the high-level discrete-adjoint methodology of [16] and implemented using `Pyadjoint` [20]. It records the forward operations on a tape and traverses them in reverse to generate the corresponding discrete adjoint. This removes the need to write an explicit reverse-time solver.

The current implementation retains the complete forward trajectory in memory. For longer simulations, checkpointing could store only selected forward states and recompute intermediate states during the reverse traversal, reducing memory consumption at the cost of additional forward computation.

2.6.2 Residual Localisation

The primal residual is localised using a hierarchical bubble–cone recovery. Firedrake’s scalar Bubble (B) and FacetBubble (FB) element families provide the auxiliary cell-interior and facet-supported test functions, respectively [18]. Temporal integration is performed using Gauss–Legendre quadrature, while the spatial forms are assembled by Firedrake using an explicitly selected quadrature degree.

2.6.3 Transfer on Nonmatching Slab Meshes

Following the refinement strategy in Section 2.4, consecutive time slabs may use different spatial meshes and finite-element spaces. The outgoing primal trace from I_{n-1} must therefore be transferred into V_n . We denote the resulting linear interpolation operator by

$$P_{n-1} : V_{n-1} \longrightarrow V_n \quad (2.60)$$

and define the transferred dG interface defect by

$$\delta_{n-1}(U_h) = U_h(t_{n-1}^+) - P_{n-1}U_h(t_{n-1}^-). \quad (2.61)$$

The reverse propagation is determined by the same discrete forward operator. Let $P_{n-1}^* : V_n \rightarrow V_{n-1}$ denote its adjoint with respect to the time-mass pairings:

$$(P_{n-1}w, z_n)_{M_n} = (w, P_{n-1}^*z_n)_{M_{n-1}} \quad \forall w \in V_{n-1}, \quad z_n \in V_n. \quad (2.62)$$

In coefficient form,

$$P_{n-1}^* = M_{n-1}^{-1}P_{n-1}^TM_n. \quad (2.63)$$

In the implementation, Firedrake assembles P_{n-1} as a cross-mesh finite-element interpolation operator. The forward trace is obtained by applying this matrix, while the reverse transfer is evaluated by a target mass action, a transpose matrix action, and a source mass solve as stated in (2.63).

Again, using pyadjoint could make life easier..? as there will be no need to include P_{n-1}^* .

Chapter 3

Experiments

3.1 Two-dimensional Rotating-Hill Heat Equation

This section shows the complete DWR adaptive workflow for a time-dependent two-dimensional heat equation. The example is a manufactured benchmark with a localised temperature peak that moves around the domain. It is therefore well suited to testing whether a space-time error indicator identifies both the spatial location of the peak and the time intervals in which an accurate representation is required.

We compare with two goal-oriented adaptive strategies: one based on a manually derived residual localisation and one based on the automated bubble-cone recovery developed in Section 2.3.2.

3.1.1 Manufactured Problem and Quantity of Interest

Let $\Omega = (0, 1)^2$, let $T = 1$, and write $Q := \Omega \times (0, T]$. We consider the heat equation

$$\partial_t u - \Delta u = f \quad \text{in } Q, \quad u = u_{\text{ex}} \quad \text{on } \partial\Omega \times (0, T], \quad u(\cdot, 0) = u_{\text{ex}}(\cdot, 0). \quad (3.1)$$

The source term f is chosen by the method of manufactured solutions so that the exact solution is

$$u_{\text{ex}}(x, y, t) = \frac{1}{1 + \alpha r^2(x, y, t)}, \quad r^2(x, y, t) = (x - c_x(t))^2 + (y - c_y(t))^2, \quad (3.2)$$

where $\alpha = 50$ and

$$(c_x(t), c_y(t)) = \left(\frac{1}{2} + \frac{1}{4} \cos(2\pi t), \frac{1}{2} + \frac{1}{4} \sin(2\pi t) \right). \quad (3.3)$$

Thus u_{ex} is a smooth, localised hill whose centre travels once around a circle of radius $1/4$ about the centre of the unit square. Although the solution remains smooth, its strongest spatial variation is concentrated near the moving peak.

The reference rotating-hill benchmark uses the space–time L^2 error to measure how well the local behaviour of the solution is captured [28]. Here we use its squared, differentiable counterpart. More precisely, we define the manufactured-error or space–time tracking functional

$$J(v) := \frac{1}{2} \int_0^T \int_{\Omega} (v - u_{\text{ex}})^2 \, dx \, dt. \quad (3.4)$$

For the numerical solution u_h ,

$$J(u_h) = \frac{1}{2} \|u_h - u_{\text{ex}}\|_{L^2(Q)}^2. \quad (3.5)$$

This quantity measures the cumulative mean-square discrepancy between the computed and exact temperature fields over the whole space–time cylinder. It penalises errors in the position, height, and shape of the moving peak, as well as errors accumulated during its motion. It is therefore an appropriate diagnostic goal for assessing space–time mesh adaptation in a manufactured-solution experiment.

Since

$$J'(u_h)(v) = \int_0^T \int_{\Omega} (u_h - u_{\text{ex}})v \, dx \, dt. \quad (3.6)$$

Accordingly, the continuous adjoint used for verification is

$$-\partial_t z - \Delta z = u_h - u_{\text{ex}} \quad \text{in } Q, \quad z = 0 \quad \text{on } \partial\Omega \times (0, T), \quad z(\cdot, T) = 0. \quad (3.7)$$

The homogeneous adjoint boundary condition follows because variations of the primal solution vanish on the strongly imposed inhomogeneous Dirichlet boundary. The terminal condition arises from integration by parts in time: the primal problem is an initial-value problem, whereas its adjoint is solved backwards from T .

3.1.2 Two Refinement Indicators

Strong-residual marking score.[4, 6] For comparison, we construct a positive residual-based activity using the classical strong residual representation for the heat equation. On a space–time cell $Q_{K,n} = K \times I_n$, define

$$R_{K,n} := f + \nabla \cdot (\kappa \nabla u_h) - \partial_t u_h, \quad (3.8)$$

and, on an interior spatial facet $F \subset \partial K$,

$$r_{K,F,n} := \frac{1}{2} \llbracket \kappa \nabla u_h \cdot n \rrbracket. \quad (3.9)$$

The factor assigns half of the flux jump to each adjacent cell. For the strongly imposed Dirichlet condition in eq. (3.1), no additional boundary-facet residual is present.

The starting point is the local strong residual action

$$\rho_{K,n}(v) = (R_{K,n}, v)_{K \times I_n} - \sum_{F \subset \partial K} (r_{K,F,n}, v)_{F \times I_n} - ([u_h]_{n-1}, v(t_{n-1}^+))_K, \quad (3.10)$$

where the last term is the dG time-interface contribution. Pairing this action with the dual error and applying the Cauchy–Schwarz inequality separately to the volume, spatial-facet, and temporal-interface terms gives

$$|(R_{K,n}, z^*)_{K \times I_n}| \leq \|R_{K,n}\|_{K \times I_n} \|z^*\|_{K \times I_n}, \quad (3.11a)$$

$$|(r_{K,F,n}, z^*)_{F \times I_n}| \leq \|r_{K,F,n}\|_{F \times I_n} \|z^*\|_{F \times I_n}, \quad (3.11b)$$

$$|([u_h]_{n-1}, z^*(t_{n-1}^+))_K| \leq \|[u_h]_{n-1}\|_K \|z^*(t_{n-1}^+)\|_K. \quad (3.11c)$$

To combine volume and facet quantities with compatible scaling, we insert the factors $h_K^{-1/2} h_K^{1/2} = 1$. Likewise, $k_n^{-1/2} k_n^{1/2} = 1$ gives the natural scaling for the temporal jump. This yields

$$\rho_{K,n}^1 := \|R_{K,n}\|_{K \times I_n} + h_K^{-1/2} \|r_{K,F,n}\|_{\partial K \times I_n}, \quad (3.12a)$$

$$\zeta_{K,n}^1 := \|z^*\|_{K \times I_n} + h_K^{1/2} \|z^*\|_{\partial K \times I_n}, \quad (3.12b)$$

$$\rho_{K,n}^2 := k_n^{-1/2} \|[u_h]_{n-1}\|_K, \quad (3.12c)$$

$$\zeta_{K,n}^2 := k_n^{1/2} \|z^*(t_{n-1}^+)\|_K. \quad (3.12d)$$

The resulting non-negative marking activity is

$$m_{K,n}^{\text{str}} := \rho_{K,n}^1 \zeta_{K,n}^1 + \rho_{K,n}^2 \zeta_{K,n}^2. \quad (3.13)$$

Bubble–recovery localisation. The automated method instead reconstructs the weak residual directly from its action on local bubble and cone test functions. For the present CG(1) $\times$ DG(0) heat discretisation, the recovered residual has three relevant supporting entities: a volume contribution on $K \times I_n$, a spatial-facet contribution on $F \times I_n$, and a left temporal-face contribution on $K \times \{t_{n-1}\}$. Their cell–slab contribution is

$$\eta_{K,n} = \eta_{\text{vol},K,n} + \eta_{\text{space},K,n} + \eta_{\text{time},K,n}. \quad (3.14)$$

The quantity used for marking is

$$m_{K,n}^{\text{bubble}} := |\eta_{K,n}|. \quad (3.15)$$

3.1.3 Results and Interpretation

The manufactured moving-hill problem was solved on $\Omega = (0, 1)^2$ over $[0, 1]$, with initial mesh parameter $n_x = n_y = 8$, $N_t = 8$ initial time slabs, and CG(1)dG(0) primal and CG(2)dG(1) enriched spaces. All estimators used the same quadrature rule (seven time quadrature points) and the nonlinear DWR identity for the quadratic running goal (3.5). Thus, though the heat equation is linear, the estimator accounts for the nonlinearity of the goal functional. The strong-residual bound and the bubble localisation both used global bulk marking with $\theta = 0.4$, the within-slab trigger 10%, and independent slabwise mesh refinement. Both adaptive sequences have nine reported levels.

method	it.	N_{st}	N_{cell}	N_t	$J(u_{\text{ex}}) - J(u_h)$	η_{global}	I_{eff}
strong	0	688	1,104	8	$-1.638\text{e-}3$	$-1.517\text{e-}3$	0.9262
strong	2	1,036	1,800	8	$-1.512\text{e-}3$	$-1.397\text{e-}3$	0.9241
strong	4	7,444	14,204	20	$-3.579\text{e-}4$	$-3.504\text{e-}4$	0.9789
strong	6	45,923	90,258	43	$-8.678\text{e-}5$	$-8.627\text{e-}5$	0.9941
strong	8	295,517	587,271	90	$-2.079\text{e-}5$	$-2.075\text{e-}5$	0.9982
bubble	0	688	1,104	8	$-1.638\text{e-}3$	$-1.517\text{e-}3$	0.9262
bubble	2	1,281	2,290	8	$-1.492\text{e-}3$	$-1.378\text{e-}3$	0.9236
bubble	4	9,008	17,366	18	$-3.852\text{e-}4$	$-3.766\text{e-}4$	0.9776
bubble	6	59,663	117,833	36	$-1.005\text{e-}4$	$-9.980\text{e-}5$	0.9934
bubble	8	401,953	800,335	72	$-2.541\text{e-}5$	$-2.536\text{e-}5$	0.9980

Table 3.1: Selected adaptive iterations for the strong-residual bound and bubble localisation. Here $N_{\text{st}} = \sum_{n=1}^{N_t} \dim(\text{CG}(1; \mathcal{T}_n))$, $N_{\text{cell}} = \sum_{n=1}^{N_t} \#\mathcal{T}_n$.

Table 3.1 shows two distinct phases. From iteration 0 to iteration 2, both methods retain $N_t = 8$ time slabs. The refinement is predominantly spatial. After temporal marking becomes active, the error decreases much more rapidly. For the strong-residual run, N_t grows from 8 at iteration 2 to 43 at iteration 6, at which point the goal error is below 10^{-4} . The bubble run increases from 8 to 36 slabs over the same iterations and reaches 1.005×10^{-4} . At the final reported iteration, the strong and bubble methods attain goal errors of 2.079×10^{-5} and 2.541×10^{-5} , respectively.

Both effectivity indices are initially approximately 0.926, and approach one monotonically after time refinement becomes active. The convergence of I_{eff} to one confirms that the DWR identity correctly estimates the quadratic goal error.

At iteration 8, the strong-residual run uses 90 time slabs and 2.96×10^5 slabwise primal DoFs, whereas the bubble run uses 72 time slabs and 4.02×10^5 slabwise primal DoFs. Thus, for the present parameters, the strong-residual strategy attains a slightly smaller goal error with fewer slabwise unknowns. These counts also indicate a different allocation of the refinement budget. The strong-residual method performs more temporal refinement, but retains fewer spatial unknowns per slab on average, suggesting that its spatial refinement is concentrated in smaller regions. This interpretation is consistent with the mesh snapshots below, in which the strong-residual meshes exhibit compact refined patches around the moving hill, whereas the bubble meshes show a more spatially distributed refinement pattern.

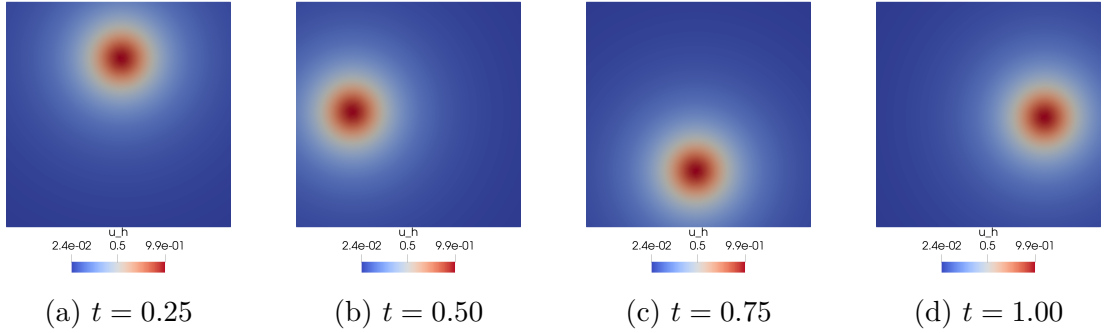


Figure 3.1: Numerical rotating-hill solution at the four physical snapshot times $t = 0.25, 0.50, 0.75, 1.00$.

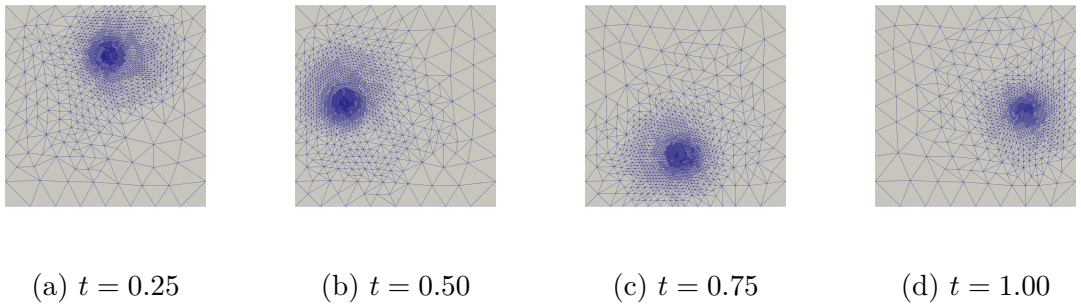


Figure 3.2: Slabwise spatial meshes from strong-residual-bound marking at $t = 0.25, 0.50, 0.75, 1.00$ at the 8th iteration.

For both localisation methods, the refined spatial regions follow the moving hill. The strong-residual indicator is a positive upper-bound quantity constructed from residual and dual-weight norms. Under the present marking rule, all space-time cell

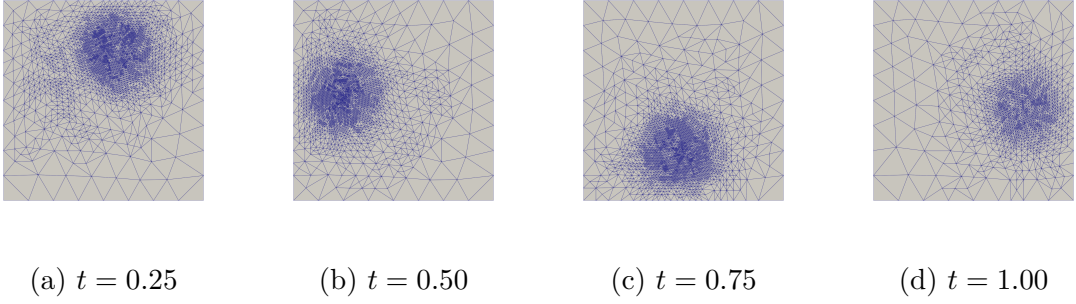


Figure 3.3: Slabwise spatial meshes from the bubble run at the physical snapshot times $t = 0.25, 0.50, 0.75, 1.00$ at the 8th iteration.

indicators are sorted in descending order and the largest entries are selected until their cumulative sum reaches $\theta = 0.4$ of the total indicator activity. The strong-residual activity is concentrated in fewer large entries and therefore produces fewer, more compact spatially refined regions. The bubble localisation combines signed recovered volume, spatial-trace and temporal-trace contributions. These components may partially cancel in the final cell indicator. Moreover, recovered facet contributions are assigned to neighbouring cells, which can distribute the indicator activity over a wider spatial region. This could explain why more cells are selected and why the bubble meshes display broader refined patches around the moving-hill trajectory in this experiment.

The results motivate a componentwise marking indicator,

$$m_{K,n}^{\text{comp}} = |\eta_{\text{vol}}| + |\eta_{\text{spatial}}| + |\eta_{\text{temporal}}| + |\eta_{\text{mixed}}|.$$

This modification prevents large entity contributions from being hidden by local sign cancellation and may therefore provide a more robust refinement indicator in cases where the recovered components have competing signs. However, it may also mark unexpected regions and increase computational cost, so its benefit must be assessed experimentally rather than assumed.

3.2 One-dimensional Advection-Diffusion Equation

The preceding heat-equation experiments examined the behaviour of the localisation procedure for a fixed observation functional. We now consider a one-dimensional advection–diffusion problem for two related purposes. First, the one-dimensional spatial domain permits the adaptive mesh to be displayed directly in the (x, t) plane,

so that spatial and temporal refinement can be observed simultaneously. Second, by keeping the primal problem and localisation procedure fixed while varying the location of the observation sensor, we isolate the influence of the goal functional, and hence of the associated dual weight, on the resulting refinement pattern.

3.2.1 Moving-pulse Problem

We consider the advection–diffusion equation

$$\partial_t u - \varepsilon \partial_{xx} u + b \partial_x u = f \quad \text{in } (0, 1) \times (0, T), \quad (3.16)$$

subject to homogeneous Dirichlet boundary conditions

$$u(0, t) = u(1, t) = 0, \quad t \in (0, T], \quad (3.17)$$

and the initial condition

$$u(x, 0) = x(1 - x) \exp[-\beta(x - x_0)^2]. \quad (3.18)$$

The forcing term f is manufactured so that the exact solution is

$$u(x, t) = x(1 - x) \exp[-\beta(x - x_0 - bt)^2]. \quad (3.19)$$

The parameters used throughout the experiments are

$$\varepsilon = 0.02, \quad b = 0.6, \quad \beta = 100, \quad x_0 = 0.2, \quad T = 1. \quad (3.20)$$

The solution represents a narrow concentration pulse transported from left to right while undergoing diffusion. Its nominal centre follows the trajectory

$$x_c(t) = x_0 + bt = 0.2 + 0.6t \quad (3.21)$$

and therefore reaches $x_c(T) = 0.8$ at the final time. The factor $x(1 - x)$ ensures compatibility with the homogeneous boundary conditions and causes the amplitude of the pulse to vary as it moves through the domain. Figure 3.4 shows the resulting space–time solution.

3.2.2 Quantity of Interest and Adjoint Sensitivity

To study the influence of the goal functional, we introduce a family of finite-width terminal detectors,

$$J_{x_d}(u) = \int_0^1 \psi_{x_d}(x) u(x, T) dx, \quad \psi_{x_d}(x) = \exp\left[-\left(\frac{x - x_d}{r_d}\right)^2\right], \quad (3.22)$$

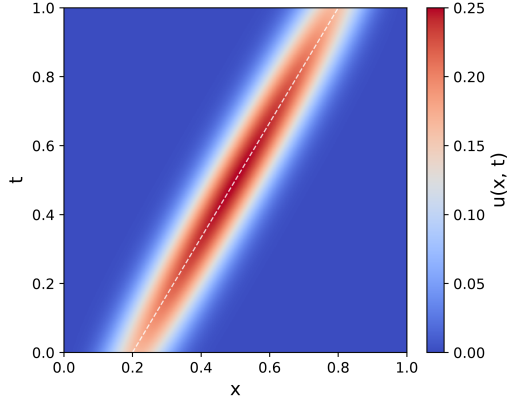


Figure 3.4: Exact solution of the 1D advection–diffusion problem. The dashed line indicates the nominal pulse trajectory $x = x_0 + bt$.

where

$$r_d = 0.05, \quad x_d \in \{0.7, 0.8, 0.9\}. \quad (3.23)$$

The functional may be interpreted as the response of an optical or concentration sensor whose sensitivity is maximal at x_d and decreases smoothly away from its centre.

The three detector locations probe physically distinct parts of the pulse at the final time. The detector at $x_d = 0.8$ is aligned with the pulse centre, the detector at $x_d = 0.7$ observes its trailing side, and the detector at $x_d = 0.9$ observes its leading side. Although the primal equation is identical in all three experiments, these observations assign different importance to errors in different regions of space–time.

Each J_{x_d} is linear in the state u . The corresponding formal adjoint problem is

$$-\partial_t z - \varepsilon \partial_{xx} z - b \partial_x z = 0 \quad \text{in } (0, 1) \times (0, T), \quad (3.24)$$

with

$$z(0, t) = z(1, t) = 0 \quad (3.25)$$

and terminal datum

$$z(x, T) = J'_{x_d}(u)(x, T) = \psi_{x_d}(x). \quad (3.26)$$

Changing the sensor centre therefore changes the terminal condition of the adjoint. The terminal datum is most sharply localised at $t = T$. To understand how this sensitivity is propagated to earlier times, we introduce the reversed time variable

$$\tau = T - t.$$

The adjoint equation then becomes

$$\partial_\tau z - \varepsilon \partial_{xx} z - b \partial_x z = 0, \quad z(x, 0) = \psi_{x_d}(x). \quad (3.27)$$

Thus, when propagated backwards in physical time, the terminal Gaussian is transported to the left by the adjoint dynamics and simultaneously broadened by diffusion.

We (therefore) expect the influence of the detector location to be most clearly visible near the terminal time, where the three dual weights remain narrow and spatially separated. Further backwards in time, diffusion broadens the weights and increases their overlap. Since the primal residual is common to all three experiments, the resulting refinement patterns are expected to become more similar at earlier times while separating towards the three observation locations near $t = T$. This behaviour will be examined in the adapted space–time meshes below.

3.2.3 One-dimensional Residual Localisation

The residual localisation requires a minor modification in one spatial dimension. In higher dimensions, the localised residual generally contains cell-interior contributions and contributions supported on spatial facets. In one dimension, however, the facets of an interval are its two endpoints. The facet contribution therefore remains present, but is represented by point evaluations at the left and right endpoints rather than by integration over edges or faces of positive dimension.

On each space–time cell

$$K \times I_n, \quad I_n = (t_{n-1}, t_n],$$

the local indicator is decomposed as

$$\eta_{K,n} = \eta_{K,n}^{\text{vol}} + \eta_{K,n}^{\text{end}} + \eta_{K,n}^{\text{time}}, \quad (3.28)$$

where $\eta_{K,n}^{\text{vol}}$ is the cell-interior contribution, $\eta_{K,n}^{\text{end}}$ contains the spatial endpoint contributions, and $\eta_{K,n}^{\text{time}}$ represents the residual at the left temporal interface of the slab.

3.2.4 Results and Interpretation

In this numerical test, the initial discretisation contains 12 spatial cells and 8 time slabs. The primal solution and numerical dual use CG1 in space and dG0 in time. The enriched dual uses the corresponding higher-order space–time approximation. A marking parameter $\theta = 0.6$ is used. A time slab is bisected when the fraction of its marked spatial cells is at least 15%.

First of all, Figure 3.5 shows the evolution of the space–time mesh for the central sensor at $x_d = 0.8$. Refinement is concentrated around the transported pulse and occurs in both the spatial and temporal directions.

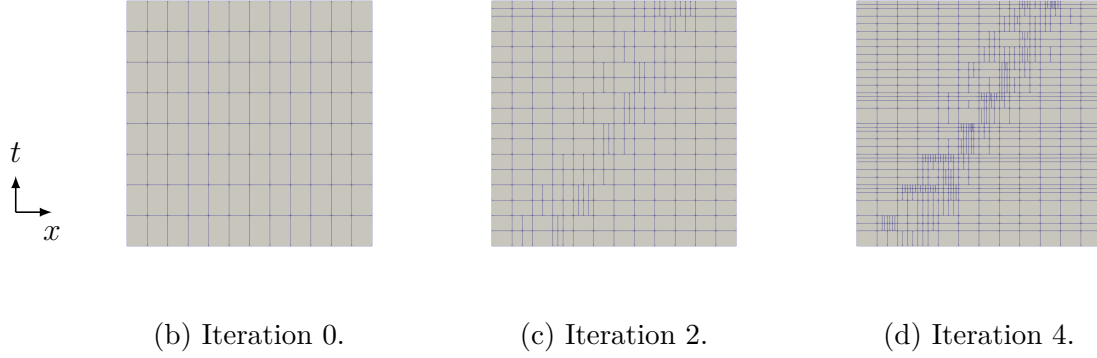


Figure 3.5: Evolution of the space–time mesh for the Gaussian detector centred at $x_d = 0.8$.

Figure 3.6 compares the space–time cells selected during the first adaptive iteration, with space shown horizontally and time increasing vertically. Although all three calculations start from the same uniform mesh, their marked cells form different bands around the trajectory of the transported pulse.

For the detector at $x_d = 0.7$, the marked cells form a relatively continuous diagonal band towards the trailing side of the pulse. Fifteen space–time cells are selected, distributed over eight time slabs. For the central detector at $x_d = 0.8$, the marked set is more concentrated: the bulk marking criterion is satisfied using only ten cells, and only two time slabs satisfy the temporal refinement criterion. In contrast, the detector at $x_d = 0.9$ produces a broader marked region at later times, shifted towards the leading side of the pulse. Sixteen cells are selected and four time slabs are marked for temporal refinement.

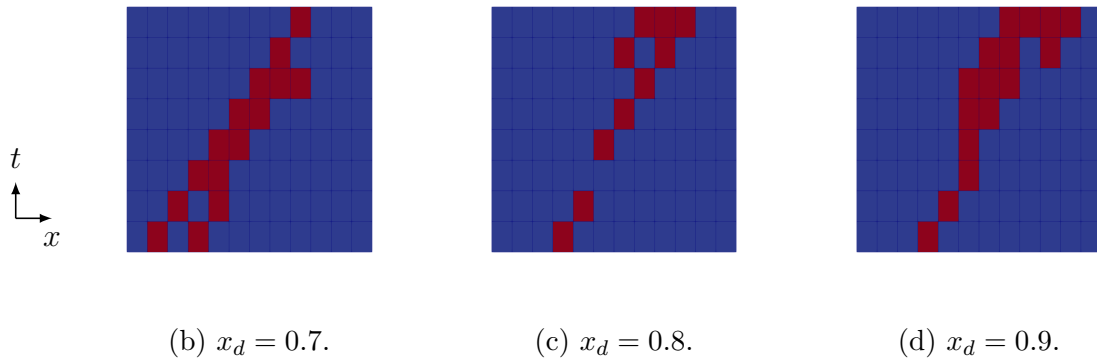


Figure 3.6: Space–time cells marked during the first adaptive iteration for Gaussian terminal sensors centred at three different locations.

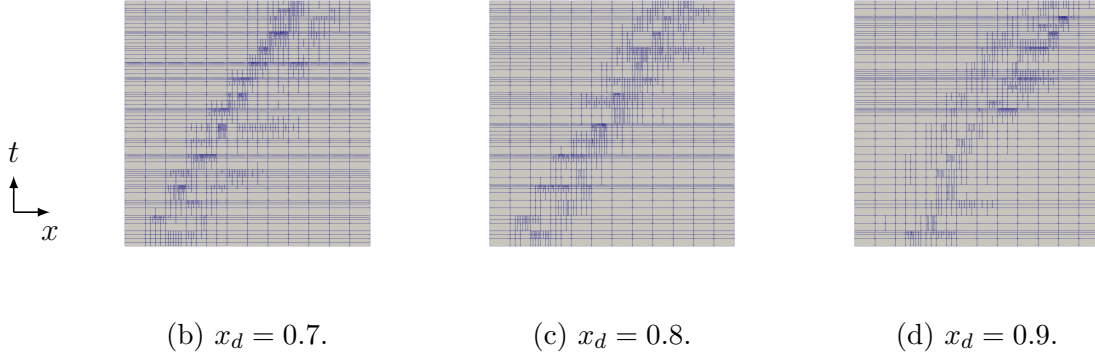


Figure 3.7: Final adapted space–time meshes for Gaussian terminal sensors centred at $x_d = 0.7$, 0.8 , and 0.9 .

The final adapted meshes are compared in Figure 3.7. Changing the detector location modifies the shape and terminal destination of the refined space–time tube. For $x_d = 0.7$, refinement follows a comparatively continuous diagonal path towards the trailing part of the pulse. For $x_d = 0.8$, the refined region remains concentrated around the central pulse trajectory as shown in Figure 3.4. For $x_d = 0.9$, the late-time refinement is shifted further to the right and bends towards the leading part of the pulse. When traced backwards in time, the three refined regions return towards the common initial pulse location near $x_0 = 0.2$. This demonstrates that the terminal sensor affects through the backward propagation of the dual weight.

3.3 Benjamin–Bona–Mahony (BBM) Equation

In this section, we turn to a 1D nonlinear BBM experiment. It tests the automated space–time localisation for a nonlinear dispersive equation, for which the nonlinear DWR identity is required. Moreover, it tests whether the entity recovery remains complete when the time derivative is paired through a spatial H^1 bilinear form rather than the spatial L^2 pairing used in the previous heat and advection–diffusion examples.

3.3.1 Primal Problem and Solitary Wave

Let $\Omega = (0, L)$ and impose periodic boundary conditions by identifying $x = 0$ with $x = L$. The spatial domain is the one-dimensional torus

$$\Omega = \mathbb{T}_L := \mathbb{R}/(L\mathbb{Z}).$$

The torus notation records that the two endpoints are the same physical point: there is no exterior spatial boundary, and the periodic seam is treated as an interior facet. The unforced BBM equation is

$$u_t - u_{xxt} + u_x + uu_x = 0 \quad \text{in } \Omega \times (0, T], \quad (3.29)$$

or, equivalently,

$$(I - \partial_{xx})u_t + \partial_x \left(u + \frac{1}{2}u^2 \right) = 0. \quad (3.30)$$

The spatial bilinear form associated with the time derivative is

$$m_{\text{BBM}}(w, v) := (w, v)_\Omega + (\partial_x w, \partial_x v)_\Omega, \quad (3.31)$$

which is an H^1 -type spatial mass pairing.

For $c \in (0, 1)$, BBM admits the solitary-wave solution on the real line

$$u_{\text{sol}}(x, t) = \frac{3c^2}{1 - c^2} \text{sech}^2 \left[\frac{c}{2} \left(x - x_0 - \frac{t}{1 - c^2} \right) \right]. \quad (3.32)$$

We take $L = 100$, $c = 0.5$, and $x_0 = 30$. The amplitude is one and the propagation speed is

$$s = \frac{1}{1 - c^2} = \frac{4}{3}.$$

Thus the wave centre moves from $x_0 = 30$ to $x_0 + sT = 54$ at $T = 18$. Throughout this interval the wave remains far from the periodic seam.

3.3.2 Quantity of Interest

We take the terminal quantity of interest

$$J(u) = (\psi, u(T))_\Omega, \quad (3.33)$$

$$\psi(x) = \exp \left[- \left(\frac{x - x_s}{r_s} \right)^2 \right], \quad x_s = 54, \quad r_s = 1. \quad (3.34)$$

It measures the wave in a narrow neighbourhood of its predicted terminal position and is therefore sensitive to phase, amplitude, and shape errors near the sensor. High-order quadrature of the solitary wave gives

$$J_{\text{ref}} = J(u_{\text{sol}}) = 1.7203. \quad (3.35)$$

3.3.3 Localisation: Mixed Space–Time Ridge

This subsection identifies the additional support entity. The derivation is used only to interpret the weak functional: the implementation continues to evaluate and project the original weak residual and does not insert a case-specific strong-residual formula.

Let

$$j_n = u_h(t_{n-1}^+) - P_n u_h(t_{n-1}^-) \quad (3.36)$$

be the dG interface defect, including transfer P_n from the preceding slab when consecutive slabs use different spatial meshes. For the spatial L^2 mass pairing used by the heat and standard advection–diffusion problems, the dG jump contributes

$$\rho_{n,L^2}^{\text{tf}}(v) = -(j_n, v(t_{n-1}^+))_{\Omega}. \quad (3.37)$$

Thus the jump itself acts only on the value trace over the temporal face.

For BBM the same defect is paired through eq. (3.31):

$$\rho_n^{\text{tf}}(v) = - \sum_{K \in \mathcal{T}_n} [(j_n, v)_K + (\partial_x j_n, \partial_x v)_K]. \quad (3.38)$$

The derivative term acts over the temporal face in the original weak form. Its cellwise distributional interpretation is

$$\rho_n^{\text{tf}}(v) = \sum_{K \in \mathcal{T}_n} (-j_n + \partial_{xx} j_n, v)_K - \sum_{F \in \mathcal{F}_n^{\text{int}}} ([\partial_x j_n], v)_F. \quad (3.39)$$

The first sum is supported on $K \times \{t_{n-1}\}$, the interior of a temporal face. The second is supported on

$$F \times \{t_{n-1}\}, \quad (3.40)$$

the intersection of a spatial facet and the temporal interface. This is the mixed space–time ridge. In one spatial dimension a facet F is a point, so the mixed ridge appears as a zero-dimensional space–time corner in the (x, t) diagram.

For the present CG1 discretisation, j_n is piecewise linear. Therefore

$$\partial_{xx} j_n = 0 \quad \text{inside each cell,} \quad [\partial_x j_n]_F \neq 0 \quad \text{in general.}$$

The derivative part of the interface defect can consequently have a substantial mixed-ridge component.

3.3.4 Results and Interpretation

The global estimator uses the nonlinear DWR identity with the unknown cubic remainder omitted.

$$\eta_{\text{global}} = \frac{1}{2}\rho(u_h)(z_h^+ - z_h) + \frac{1}{2}\rho^*(u_h, z_h)(u_h^+ - u_h) - \rho(u_h)(z_h). \quad (3.41)$$

The low-order and enriched spaces are, respectively,

$$V_h^{\text{low}} = \text{CG}(1)_x \otimes \text{DG}(0)_t, \quad V_h^+ = \text{CG}(2)_x \otimes \text{DG}(1)_t.$$

We take the bulk marking fraction as 0.35 and bisect a time slab when at least 10% of its spatial cells are selected. We have initially 50 spatial cells and 6 time slabs. The terminal time is set to be 18.

To verify algebraic completeness as stated in the previous subsection 3.3.3, each adaptive iterate is post-processed on the same primal solution, adjoint solution, and mesh in two ways: with all recovered entities and after deleting only the signed mixed-ridge contribution. Define

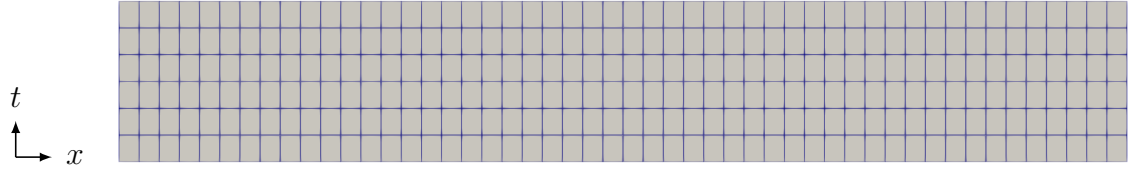
$$\delta_{\text{loc}} := \sum_{n,K} \eta_{K,n} - \eta_{\text{global}}. \quad (3.42)$$

With the mixed ridge included, $|\delta_{\text{loc}}|$ stays at floating-point roundoff. This test in Table 3.2 demonstrates that the mixed ridge term is required to reproduce the global DWR estimator for this BBM calculation.

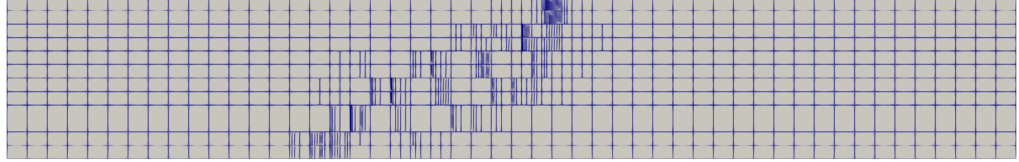
Iteration	Space-time cells	$ \delta_{\text{loc}} $	$ \delta_{\text{loc}}^{\text{no mix}} $	$\frac{ \delta_{\text{loc}}^{\text{no mix}} }{ \eta_{\text{global}} }$
0	300	4.44×10^{-16}	2.85×10^{-2}	3.33%
4	350	5.55×10^{-16}	3.07×10^{-2}	3.56%
8	856	9.99×10^{-16}	5.11×10^{-2}	5.65%
12	2816	2.11×10^{-15}	6.47×10^{-2}	8.78%

Table 3.2: Mixed-ridge ablation performed on identical discrete primal and adjoint solutions.

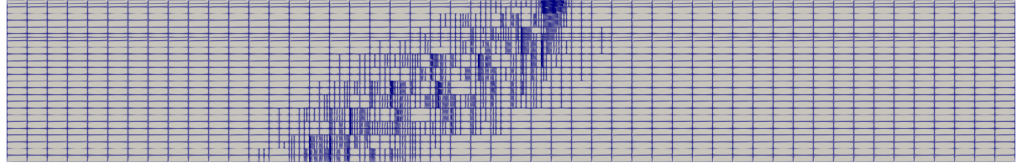
Figure 3.8 displays the adapted meshes in the (x, t) plane. The initial mesh is uniform. By iteration 8, local spatial and temporal refinements form a recognisable diagonal corridor from the initial wave position towards the terminal sensor. At



(b) Iteration 0: 300 cells.



(c) Iteration 8: 856 cells.



(d) Iteration 12: 2816 cells.

Figure 3.8: Evolution of the independently refined slabwise space-time mesh.

iteration 12 this corridor is substantially denser, while cells far from it remain comparatively coarse. Its alignment with the exact space-time wave whose centre moves from $x = 30$ at $t = 0$ to $x = 54$ at $T = 18$.

Finally, selected iterations are listed in table 3.3. The goal error remains appreciable because a narrow nonlinear wave is transported over a long time interval on relatively modest meshes. The main estimator result is the improvement in effectivity from 0.7306 to 0.9956.

Iter.	Cells	N_t	Terminal DOFs	$J(u_h)$	$ J_{\text{ref}} - J(u_h) $	I_{eff}
0	300	6	50	0.548244	1.172006	0.730579
6	465	7	68	0.576199	1.144052	0.764990
8	856	11	82	0.704237	1.016014	0.890155
10	1372	15	106	0.787125	0.933125	0.939198
12	2816	26	123	0.980816	0.739434	0.995630

Table 3.3: Selected iterations of the periodic BBM adaptive calculation.

3.4 Parabolic p -Laplace Problem on an L-shaped Domain

The preceding experiments considered smooth spatial settings and were useful for verifying the automated residual localisation. They did not yet provide a comparison with uniform refinement. We therefore consider the classical L-shaped domain benchmark based on [14, Section 6.4], together with the regularised parabolic p -Laplace model analysed in [12]. The example combines reduced primal regularity with goal functionals that may or may not be sensitive to the singularity.

3.4.1 p -Laplace Problem and Goal Functional

Let

$$\Omega = (-1, 1)^2 \setminus ([0, 1] \times [-1, 0]), \quad Q = \Omega \times (0, 1). \quad (3.43)$$

We solve

$$\partial_t u - \nabla \cdot \left[(|\nabla u|^2 + \varepsilon^2)^{(p-2)/2} \nabla u \right] = f \quad \text{in } Q, \quad (3.44)$$

with homogeneous Dirichlet conditions and $u(\cdot, 0) = 0$. Throughout this section,

$$p = 4, \quad \varepsilon = 10^{-2}.$$

The regularisation removes the degeneracy of the diffusion law at $|\nabla u| = 0$ and supplies the differentiability required by the nonlinear DWR identity [12].

Writing (r, ϑ) for polar coordinates centred at the re-entrant corner, the manufactured solution is

$$u_{\text{ex}}(x, y, t) = \frac{3}{2}(1 - x^2)^2(1 - y^2)^2 r^{2/3} \sin\left(\frac{2\vartheta}{3}\right) \sin t. \quad (3.45)$$

The source term is obtained by inserting this expression into the differential equation. Following [14, Section 6.4.3], we first consider the terminal spatial functional

$$J_1(u) = \int_{\Omega} u(x, T) \, dx. \quad (3.46)$$

The exact value used in the computation is $J_1(u_{\text{ex}}) = 0.4142466902$.

We then consider the localised nonlinear goal functional

$$J_2(u) = \int_{0.5}^{0.75} \int_{\Omega_I} |\nabla u(x, t)|^4 \, dx \, dt, \quad \Omega_I = \left(-\frac{15}{16}, -\frac{11}{16}\right) \times \left(\frac{11}{16}, \frac{15}{16}\right), \quad (3.47)$$

which is nonlinear, spatially localised, and active only on the time window $[0.5, 0.75]$.

The reference value used in the computation is $J_1(u_{\text{ex}}) = 5.2366347 \times 10^{-5}$.

For the following experiments, the primal space is CG(1)–DG(0) and both the enriched primal and adjoint use CG(2)–DG(1). The nonlinear DWR identity is localised by hierarchical bubble recovery.

3.4.1.1 Global Terminal Goal and Componentwise Marking

Since the terminal spatial-average functional J_1 is global in space, it retains some sensitivity to the singularity at the re-entrant corner. The adaptive computation employs the spatial bulk parameter $\theta = 0.35$ and a temporal marking fraction of 0.10.

The computed goal error exhibits an overall decreasing trend over the six adaptive cycles, while the global effectivity starts from 0.0559 and increases to 1.0550 at iteration 7. The localised effectivity is much closer to one throughout the sequence. Figure 3.9 shows the spatial meshes associated with four representative time slabs from the final adaptive sequence. The refinement becomes progressively denser near the re-entrant corner and extends into its surrounding region as the terminal time is approached. This pattern is consistent with the global terminal functional retaining sensitivity to the reduced regularity generated by the corner singularity.

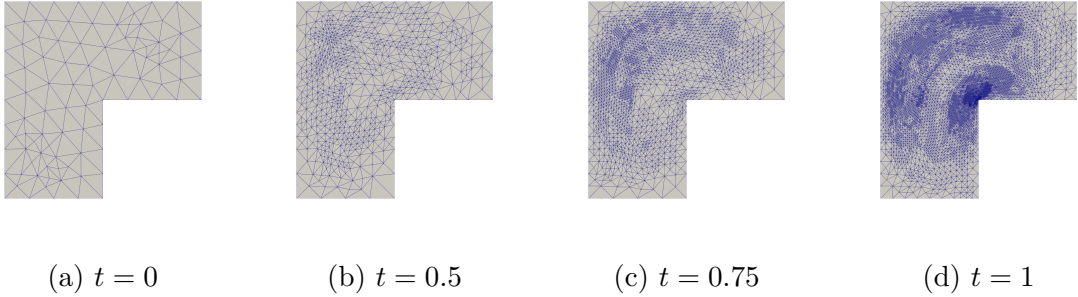


Figure 3.9: Spatial meshes at time $t = 0, 0.5, 0.75, 1$ for goal J_1 at iteration 7.

3.4.1.2 Localised Nonlinear Goal

For goal functional J_2 , the observation rectangle is represented as a separate **Netgen** material subdomain so that its boundary consists of mesh facets. The initial discretisation contains eight time slabs and uses the fitted mesh with 75 terminal spatial degrees of freedom. For the adaptive run, the spatial refinement fraction is set to be $\theta = 0.30$. A slab is bisected when at least 10% of its cells have been selected by the

global marking step. The uniform run refines every spatial cell and bisects every time slab. Both runs start from the same space–time discretisation.

Figure 3.10 compares the true J_1 error against $N_{\text{st}} = \sum_{n=1}^{N_t} \dim V_h^n$, which counts the actual slabwise primal unknowns. The errors and effectivity indices at representative iterates are reported in Table 3.4.

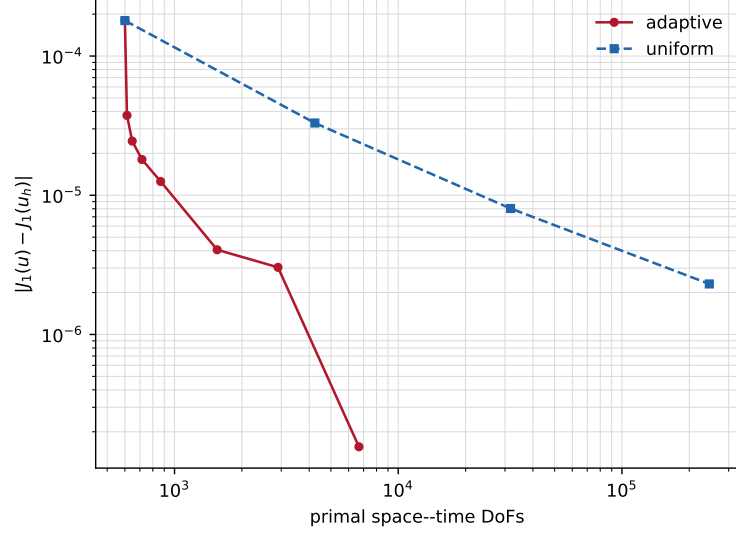


Figure 3.10: J_1 error against total primal space–time degrees of freedom for adaptive iterations 0–7 and the uniform sequence 0–3.

Table 3.4: Representative fitted- J_1 results.

strategy/iterate	N_{st}	N_t	$ e_{J_1} $	$I_{\text{eff}}^{\text{glob}}$	$I_{\text{eff}}^{\text{loc}}$
initial	600	8	1.798×10^{-4}	0.925	0.926
uniform 1	4 240	16	3.304×10^{-5}	1.082	1.063
uniform 2	31 776	32	8.036×10^{-6}	1.023	1.004
uniform 3	245 824	64	2.307×10^{-6}	1.007	0.983
adaptive 2	647	8	2.450×10^{-5}	1.140	1.106
adaptive 4	867	8	1.256×10^{-5}	1.062	0.972
adaptive 5	1 551	9	4.060×10^{-6}	1.321	1.092
adaptive 6	2 901	11	3.036×10^{-6}	0.970	0.645
adaptive 7	6 673	14	1.561×10^{-7}	0.550	4.358

The adaptive sequence reduces the error much more rapidly than uniform refinement. Uniform refinement uses approximately $245824/2901 \simeq 84.7$ times as many primal space–time degrees of freedom to reach a comparable goal accuracy.

However, from the table, an unexpected decrease to $I_{\text{eff}}^{\text{glob}} = 0.550$ occurs at adaptive iterate 7. The available reference goal value shows that this is associated with a failure of the saturation property underlying the DWR reliability and efficiency theory [14]. With u_h^+ denoting the enriched primal solution, the observed ratio

$$b_h^{\text{obs}} := \frac{|J_1(u) - J_1(u_h^+)|}{|J_1(u) - J_1(u_h)|}$$

equals $0.1296 < 1$ at iterate 6, where the enriched approximation is clearly more accurate, but increases to $1.382 > 1$ at iterate 7. At the latter level the low-order goal error has crossed zero and has the very small magnitude 1.561×10^{-7} , whereas the enriched goal lies on the opposite side of the exact value with error 2.157×10^{-7} . Enrichment no longer reduces the goal error, and the saturation-based estimate is not guaranteed. This suggested that the enriched problem should be strengthened, for example by increasing its spatial or temporal polynomial degree, if further accuracy is required.

does this make sense? though I_{eff} is incorrect but the error is indeed decreasing

Finally, Figure 3.11 and 3.12 show the actual two-dimensional mesh carried by the selected time slab.

The mesh sequence clarifies the advantage of the goal-oriented adaptivity. At $t = 0.625$ the selected slab contains 1 473 cells and at $t = 0.75$ it contains 1 959, whereas the terminal slab contains only the initial 116 cells. Refinement is concentrated around Ω_I and its surrounding sensitivity region. By contrast, the re-entrant corner is not refined to a comparable extent. This behaviour is expected for goal-oriented adaptivity as the DWR algorithm does not attempt to resolve every loss of primal regularity.

In summary, the L-shaped problem supplies the adaptive–uniform comparison that the preceding smooth tests did not stress. Here DWR-driven space-time adaptivity is more efficient than uniform refinement for the prescribed localised goal functional J_2 .

3.5 Incompressible Navier–Stokes Equations

This final experiment extends the automated space–time framework to a genuinely mixed problem: nonstationary incompressible Navier–Stokes flow past a cylinder. It follows the space–time DWR viewpoint of Besier and Rannacher [7] and the tensor-product treatment of Roth et al. [26], while retaining the bubble–cone residual localisation developed in this thesis. Galerkin time stepping and the assembly of the

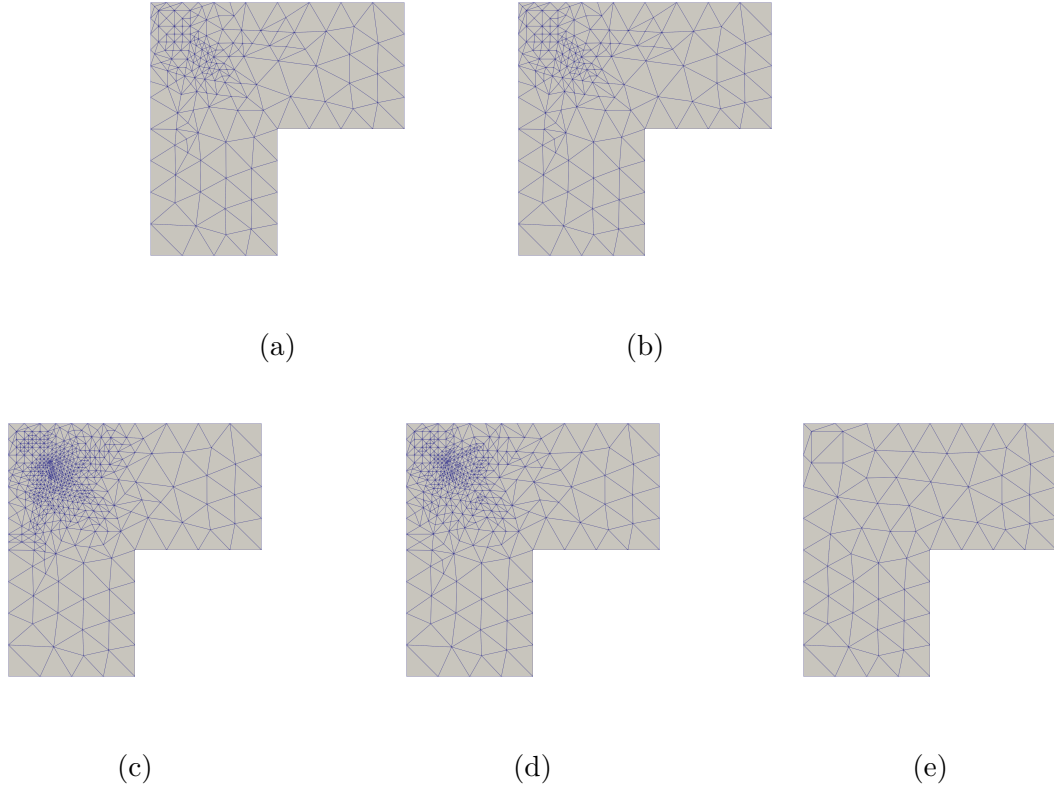


Figure 3.11: Spatial meshes for goal J_2 at iteration 7. (a) shows the initial mesh, (b)–(e) show the spatial meshes carried by the adaptive time slabs whose right endpoints are $t = 0.5, 0.625, 0.75$, and 1 , respectively.

mixed variational forms are automated in Firedrake/Irksome; the latter supports discontinuous Galerkin formulations of semidiscrete variational systems and mixed differential–algebraic equations [2].

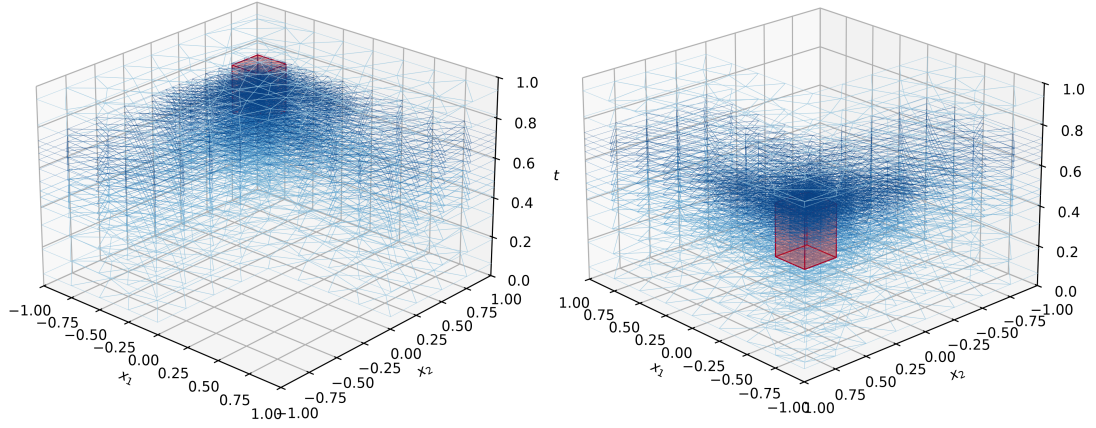


Figure 3.12: Stacked representation of the slabwise spatial meshes at adaptive iterate 7, in exterior view and a view through the re-entrant corner, respectively. The red prism represents the space-time observation region $Q_I = \Omega_I \times [0.5, 0.75]$.

Chapter 4

Conclusions and Future Work

Here we summarise the work that has been presented in this dissertation and discuss possible areas for future work.

4.1 Summary

Give a summary of what has been done. You might do this chapter by chapter but be sure to highlight all the important points.

4.2 Future Work

You don't actually have to list further work, but most people do this so it seems unusual not to. Just think what you would do next if you had more time on this project.

4.3 Conclusion

You might like to give a final conclusion so the reader is left remembering what you have done, rather than what you would do if there wasn't a submission deadline.

4.4 Page Numbering

Remember that the page count stops after the final page of your final chapter so the last page of the conclusions should be at most 54 to avoid a penalty.

References

- [1] Mark Ainsworth and J.Tinsley Oden. A posteriori error estimation in finite element analysis. *Computer Methods in Applied Mechanics and Engineering*, 142(1):1–88, 1997.
- [2] Boris D. Andrews, Pablo Brubeck, Patrick E. Farrell, Robert C. Kirby, and Scott P. MacLachlan. Automated galerkin time stepping in irksome, 2026.
- [3] I. Babuvška and W. C. Rheinboldt. Error estimates for adaptive finite element computations. *SIAM Journal on Numerical Analysis*, 15(4):736–754, 1978.
- [4] Wolfgang Bangerth and Rolf Rannacher. *Adaptive Finite Element Methods for Differential Equations*. Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2003.
- [5] Roland Becker and R. Rannacher. A feed-back approach to error control in finite element methods: Basic analysis and examples. *East-West Journal of Numerical Mathematics*, 4, 01 1996.
- [6] Roland Becker and Rolf Rannacher. An optimal control approach to a posteriori error estimation in finite element methods. *Acta Numerica*, 10:1–102, 2001.
- [7] Michael Besier and Rolf Rannacher. Goal-oriented space–time adaptivity in the finite element galerkin method for the computation of nonstationary incompressible flow. *International Journal for Numerical Methods in Fluids*, 70(9):1139–1166, 2012.
- [8] Malte Braack and Alexandre Ern. A posteriori control of modeling errors and discretization errors. *Multiscale Modeling & Simulation*, 1(2):221–238, 2003.
- [9] Eric T. Chung, Wing Tat Leung, and Sara Pollock. Goal-oriented adaptivity for gmsfem. *Journal of Computational and Applied Mathematics*, 296:625–637, 2016.

- [10] Emmanuel Creusé, Serge Nicaise, and Zuqi Tang. Goal-oriented error estimation based on equilibrated flux and potential reconstruction for the approximation of elliptic and parabolic problems. *Computers & Mathematics with Applications*, 146:323–338, 2023.
- [11] Willy Dörfler. A convergent adaptive algorithm for poisson’ s equation. *SIAM Journal on Numerical Analysis*, 33(3):1106–1124, 1996.
- [12] B. Endtmayer, U. Langer, and A. Schafelner. Goal-oriented adaptive space-time finite element methods for regularized parabolic p-laplace problems. *Computers & Mathematics with Applications*, 167:286–297, 2024.
- [13] B. Endtmayer, U. Langer, and T. Wick. Multigoal-oriented error estimates for non-linear problems, 2018.
- [14] Bernhard Endtmayer, Ulrich Langer, Thomas Richter, Andreas Schafelner, and Thomas Wick. Chapter two - a posteriori single- and multi-goal error control and adaptivity for partial differential equations. In Franz Chouly, Stéphane P.A. Bordas, Roland Becker, and Pascal Omnes, editors, *Error Control, Adaptive Discretizations, and Applications, Part 2*, volume 59 of *Advances in Applied Mechanics*, pages 19–108. Elsevier, 2024.
- [15] Kenneth Eriksson, Don Estep, Peter Hansbo, and Claes Johnson. Introduction to adaptive methods for differential equations. *Acta Numerica*, 4:105–158, 1995.
- [16] P. E. Farrell, D. A. Ham, S. W. Funke, and M. E. Rognes. Automated derivation of the adjoint of high-level transient finite element programs. *SIAM Journal on Scientific Computing*, 35(4):C369–C393, 2013.
- [17] Patrick E. Farrell, Robert C. Kirby, and Jorge Marchena-Menéndez. Irksome: Automating runge–kutta time-stepping for finite element methods. *ACM Trans. Math. Softw.*, 47(4), September 2021.
- [18] David A. Ham, Paul H. J. Kelly, Lawrence Mitchell, Colin Cotter, Rob C. Kirby, Koki Sagiyama, Nacime Bouziani, Sophia Vorderwuelbecke, Thomas Gregory, Jack Betteridge, Daniel R. Shapero, Reuben Nixon-Hill, Connor Ward, Patrick E. Farrell, Pablo D. Brubeck, India Marsden, Thomas H. Gibson, Miklós Homolya, Tianjiao Sun, Andrew T. T. McRae, Fabio Luporini, Alastair Gregory, Michael Lange, Simon W. Funke, Florian Rathgeber, Gheorghe-Teodor Bercea, and Graham R. Markall. Firedrake user manual, May 2023.

- [19] Dmitri Kuzmin and Sergey Korotov. Goal-oriented a posteriori error estimates for transport problems. *Mathematics and Computers in Simulation*, 80(8):1674–1683, 2010. ESCO 2008 Conference.
- [20] Sebastian K. Mitusch, Simon W. Funke, and Jørgen S. Dokken. dolfin-adjoint 2018.1: automated adjoints for fenics and firedrake. *Journal of Open Source Software*, 4(38):1292, 2019.
- [21] Pedro Morin, Ricardo H. Nochetto, and Kunibert G. Siebert. Convergence of adaptive finite element methods. *SIAM Review*, 44(4):631–658, 2002.
- [22] Igor Mozolevski and Serge Prudhomme. Goal-oriented error estimation based on equilibrated-flux reconstruction for finite element approximations of elliptic problems. *Computer Methods in Applied Mechanics and Engineering*, 288:127–145, 2015. Error Estimation and Adaptivity for Nonlinear and Time-Dependent Problems.
- [23] R. Rannacher and J. Vihharev. Adaptive finite element analysis of nonlinear problems: balancing of discretization and iteration errors. *Journal of Numerical Mathematics*, 21(1):23–62, 2013.
- [24] Thomas Richter and Thomas Wick. Variational localizations of the dual weighted residual estimator. *Journal of Computational and Applied Mathematics*, 279:192–208, 2015.
- [25] Marie E. Rognes and Anders Logg. Automated goal-oriented error control i: Stationary variational problems. *SIAM Journal on Scientific Computing*, 35(3):C173–C193, 2013.
- [26] Julian Roth, {Jan Philipp} Thiele, Uwe Köcher, and Thomas Wick. Tensor-product space-time goal-oriented error control and adaptivity with partition-of-unity dual-weighted residuals for nonstationary flow problems. *Computational Methods in Applied Mathematics*, 24(1):185–214, January 2024. Funding Information: The first and fourth authors acknowledge the funding of the German Research Foundation (DFG; <http://dx.doi.org/10.13039/501100001659>) within the framework of the International Research Training Group on Computational Mechanics Techniques in High Dimensions GRK 2657 under Grant Number 433082294.

- [27] Michael Schmich and Boris Vexler. Adaptivity with dynamic meshes for space-time finite element discretizations of parabolic equations. *SIAM Journal on Scientific Computing*, 30(1):369–393, 2008.
- [28] Jan Philipp Thiele and Thomas Wick. Numerical modeling and open-source implementation of variational partition-of-unity localizations of space-time dual-weighted residual estimators for parabolic problems. *Journal of Scientific Computing*, 99(1):25, 2024.